QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

librustzcash-en @ 8672a4b2d60f · Electric Coin Company (Zcash) — public repository (passive review)

Organization	QSI
Report date	2026-11-04 18:54 UTC
Report version	1
Audit run	731f2158-4b86-4486-8a1f-4d3fbb62dba0 · Document audit · 2026-11-04 18:54 UTC
Analyzer engine	qsi-engine 1.3.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`03b3c2491116445855ca64e958f067d951c66e342cfeed96f84e97c91650ed99`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

51	0	2	10	32
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	42
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	3
Improvement recommendation	5
Insufficient data	1

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-12058	High	rustls-webpki 0.101.7 is affected by GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104)	37.6
QSI-2026-12062	High	rustls-webpki 0.103.3 is affected by GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104)	37.6
QSI-2026-12067	Medium	serde_with 3.16.0 is affected by GHSA-7gcf-g7xr-8hxj	44.2
QSI-2026-12031	Medium	Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (4 file(s)) preliminary	35.6
QSI-2026-12032	Medium	Quantum-vulnerable ECDSA: transaction and message signing (15 file(s)) preliminary	35.6

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/zcash/librustzcash @ 8672a4b2d60f432664780e6a4e85f49a36805a2a — evidence links open the lines at this commit.

Supplied inputs

333 files, 8100632 bytes in total, 3 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.3.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	307	.github/helpers/check-dep-graph.py: 115 lines scanned with 16 rule patterns and 145 registry patterns; .github/helpers/check-release-tags.py: 167 lines scanned with 16 rule patterns and 145 registry patterns; components/eip681/src/error.rs: 290 lines scanned with 16 rule patterns and 145 registry patterns
dependencies	checked	21	Cargo.lock: 690 dependencies parsed (690 pinned); Cargo.toml: 101 dependencies parsed (10 pinned); components/eip681/Cargo.toml: 11 dependencies parsed (0 pinned)
secrets	checked	1	components/eip681/src/request.rs: 3 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name)
tls_code	checked	2	zcash_client_backend/src/tor.rs: TLS key-exchange settings read; zcash_client_backend/src/tor/http.rs: TLS key-exchange settings read
entropy	checked	1	random-number generation: 307 code files read with 16 macro definitions from the tree; 0 ineffective guard(s), 0 weak random-bytes provider(s), 0 weak key-generation site(s)
osv_advisories	checked	1	700 pinned packages queried at OSV.dev; 27 affected (2 of them only in development or build dependencies)

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 330 (input files: C)

Architecture overview

Static review of the source tree librustzcash-en at 8672a4b2d60f432664780e6a4e85f49a36805a2a. 333 files analysed; of 484 files, 54 are test, fuzzing, benchmark or example code and were left out, 97 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. Files no analyzer reads include: .editorconfig, .gitattributes, .github/actions/prepare/action.yml, .github/dependabot.yml, .github/workflows/aggregate-audits.yml. 120 files in build-output or dependency-cache directories (.git, .hg, .venv, __pycache__, build, dist, node_modules, target, vendor, venv) were not read. No requests were sent to the owner's systems.

Overall security posture

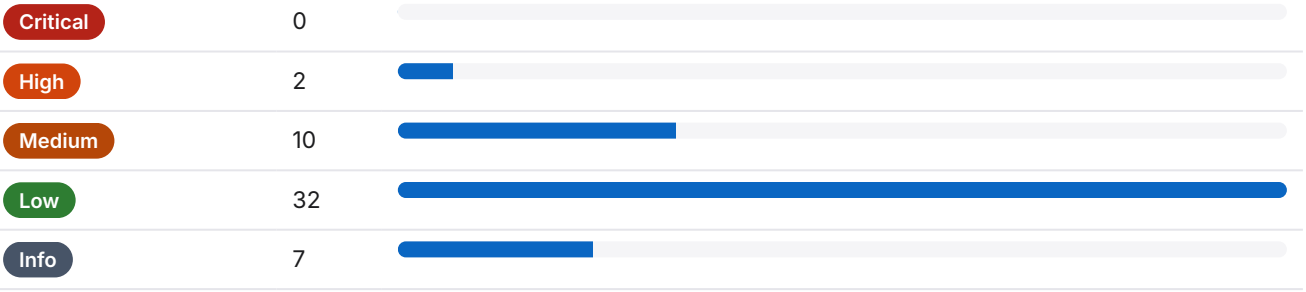
6
Indicative posture index (0–100)

17.7%
Quantum-vulnerable share of inventory

147
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-12058	rustls-webpki 0.101.7 is affected by GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104)	Known vulnerable component	High	Low
QSI-2026-12062	rustls-webpki 0.103.3 is affected by GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104)	Known vulnerable component	High	Low
QSI-2026-12067	serde_with 3.16.0 is affected by GHSA-7gcf-g7xr-8hxj	Known vulnerable component	Medium	Medium
QSI-2026-12031	Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (4 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-12032	Quantum-vulnerable ECDSA: transaction and message signing (15 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-12033	Quantum-vulnerable BLS12-381: transaction and message signing (4 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-12038	bytes 1.10.1 is affected by GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007)	Known vulnerable component	Medium	Low
QSI-2026-12049	rsa 0.9.6 is affected by GHSA-4grx-2x9w-596c (CVE-2023-49092, GHSA-c38w-74pg-36hr, RUSTSEC-2023-0071)	Known vulnerable component	Medium	Low
QSI-2026-12051	rustls 0.21.12 is affected by GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346)	Known vulnerable component	Medium	Low
QSI-2026-12054	rustls 0.23.25 is affected by GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285)	Known vulnerable component	Medium	Low
QSI-2026-12055	rustls 0.23.25 is affected by GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346)	Known vulnerable component	Medium	Low
QSI-2026-12064	rustls-webpki 0.103.3 is affected by GHSA-pwjx-qhcg-rvj4 (CVE-2026-93602, RUSTSEC-2026-0049)	Known vulnerable component	Medium	Low
QSI-2026-12047	rand 0.8.5 is affected by GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097)	Known vulnerable component	Low	Medium

QSI-2026-12048	rand 0.9.2 is affected by GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097)	Known vulnerable component	Low	Medium
QSI-2026-12050	rsa 0.9.6 is affected by GHSA-9c48-w39g-hm26 (CVE-2026-21895)	Known vulnerable component	Low	Low
QSI-2026-12052	rustls 0.21.12 is affected by GHSA-h2v8-394m-p7vv	Known vulnerable component	Low	Low
QSI-2026-12053	rustls 0.21.12 is affected by GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349)	Known vulnerable component	Low	Low
QSI-2026-12057	rustls 0.23.25 is affected by GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349)	Known vulnerable component	Low	Low
QSI-2026-12059	rustls-webpki 0.101.7 is affected by GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098)	Known vulnerable component	Low	Low
QSI-2026-12060	rustls-webpki 0.101.7 is affected by GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099)	Known vulnerable component	Low	Low
QSI-2026-12063	rustls-webpki 0.103.3 is affected by GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098)	Known vulnerable component	Low	Low
QSI-2026-12065	rustls-webpki 0.103.3 is affected by GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099)	Known vulnerable component	Low	Low
QSI-2026-12043	keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12056	rustls 0.23.25 is affected by GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12036	anyhow 1.0.100 is affected by RUSTSEC-2026-0190 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12039	crossbeam-epoch 0.9.18 is affected by RUSTSEC-2026-0204 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12040	crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12041	ctutils 0.4.2 is affected by RUSTSEC-2026-0385 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12042	event-listener 5.3.1 is affected by RUSTSEC-2026-0221 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12044	memmap2 0.9.4 is affected by RUSTSEC-2026-0186 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12061	rustls-webpki 0.101.7 is affected by RUSTSEC-2026-0382 (CVE-2026-111943) (no severity rating)	Known vulnerable component	Low	Low, preliminary

QSI-2026-12066	rustls-webpki 0.103.3 is affected by RUSTSEC-2026-0382 (CVE-2026-111943) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12069	webpki-roots 0.25.4 is affected by RUSTSEC-2026-0381 (CVE-2026-111943) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12070	webpki-roots 1.0.3 is affected by RUSTSEC-2026-0381 (CVE-2026-111943) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12071	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0395 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12072	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0397 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12073	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0399 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12074	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0400 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12075	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0401 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12076	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0402 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12077	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0403 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12078	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0404 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12079	zstd-safe 7.1.0 is affected by RUSTSEC-2026-0406 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12080	zstd-sys 2.0.10+zstd.1.5.6 is affected by RUSTSEC-2026-0410 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-12068	tokio 1.50.0 is affected by RUSTSEC-2026-0386 (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-12081	3 advisories in 2 development or build-only crates.io package(s)	Improvement recommendation	Info	Medium, preliminary
QSI-2026-12034	No post-quantum capable cryptographic library detected	Insufficient data	Info	Medium, preliminary
QSI-2026-12035	adler 1.0.2: informational advisory RUSTSEC-2025-0056 (unmaintained)	Improvement recommendation	Info	Low
QSI-2026-12037	bincode 2.0.1: informational advisory RUSTSEC-2025-0141 (unmaintained)	Improvement recommendation	Info	Low
QSI-2026-12045	paste 1.0.15: informational advisory RUSTSEC-2024-0436 (unmaintained)	Improvement recommendation	Info	Low

QSI-2026-12046	proc-macro-error2 2.0.1: informational advisory RUSTSEC-2026-0173 (unmaintained)	Improvement recommendation	Info	Low

Detailed findings

QSI-2026-12058 · DEP-KNOWN-ADVISORY

High

rustls-webpki 0.101.7 is affected by GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104) for crates.io package rustls-webpki at version 0.101.7: rustls-webpki: Denial of service via panic on malformed CRL BIT STRING. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.101.7
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	37.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-82j2-j2ch-gfr8
```

EVIDENCE

tool_output · OSV.dev GHSA-82j2-j2ch-gfr8 · manifest · SHA-256 a2f685e85eb5c808...


```
{"id": "GHSA-82j2-j2ch-gfr8", "aliases": ["CVE-2026-93599", "RUSTSEC-2026-0104"], "modified":
"2026-09-19T03:55:33.412079637Z", "summary": "rustls-webpki: Denial of service via panic on
malformed CRL BIT STRING", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/
S:U/C:N/I:N/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.103.13, 0.104.0-alpha.7).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-82j2-j2ch-gfr8

QSI-2026-12062 · DEP-KNOWN-ADVISORY

High

rustls-webpki 0.103.3 is affected by GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104) for crates.io package rustls-webpki at version 0.103.3: rustls-webpki: Denial of service via panic on malformed CRL BIT STRING. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.103.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	37.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-82j2-j2ch-gfr8
```

EVIDENCE

tool_output · OSV.dev GHSA-82j2-j2ch-gfr8 · manifest · SHA-256 a2f685e85eb5c808…


```
{"id": "GHSA-82j2-j2ch-gfr8", "aliases": ["CVE-2026-93599", "RUSTSEC-2026-0104"], "modified": "2026-09-19T03:55:33.412079637Z", "summary": "rustls-webpki: Denial of service via panic on malformed CRL BIT STRING", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.103.13, 0.104.0-alpha.7).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-82j2-j2ch-gfr8

QSI-2026-12067 · DEP-KNOWN-ADVISORY

Medium

serde_with 3.16.0 is affected by GHSA-7gcf-g7xr-8hxj

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-7gcf-g7xr-8hxj for crates.io package serde_with at version 3.16.0: serde_with: KeyValueMap serialization panics on empty sequence or map entries. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/serde_with@3.16.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	44.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-7gcf-g7xr-8hxj
```

EVIDENCE

tool_output · OSV.dev GHSA-7gcf-g7xr-8hxj · manifest · SHA-256 b42c751004bfb387…


```
{"id": "GHSA-7gcf-g7xr-8hxj", "aliases": [], "modified": "2026-09-10T03:51:10.897685062Z",
"summary": "serde_with: KeyValueMap serialization panics on empty sequence or map entries",
"severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:L/SC:N/
SI:N/SA:N"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (3.21.0).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-7gcf-g7xr-8hxj

QSI-2026-12031 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (4 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

13 reference(s) to Pedersen commitment across 4 file(s), e.g. pczt/src/orchard.rs:23. Role, judged from paths, function names and nearby code: threshold signing and MPC protocols. Paillier, Pedersen commitments and the discrete-logarithm proofs inside threshold ECDSA and EdDSA protocols all fall to Shor's algorithm; a post-quantum threshold scheme is a protocol change, not a parameter change.

Affected asset	source-code:Pedersen commitment:threshold_mpc

Data source	pczt/src/orchard.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 18:54 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open pczt/src/orchard.rs:23 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · pczt/src/orchard.rs · line 23 · SHA-256 ceab0b6e960329ba… · File 483324d3e983919c…

```
   22 |         },
   23 |         value::{NoteValue, ValueCommitTrapdoor, ValueCommitment},
   24 |     },
```

file_excerpt · pczt/src/orchard.rs · line 2161 · SHA-256 6b6748f70bf576cd… · File 483324d3e983919c…

```
 2160 |         );
 2161 |         let rcv = ValueCommitTrapdoor::from_bytes(
 2162 |             self.rcv
```

file_excerpt · pczt/src/orchard.rs · line 2168 · SHA-256 14e2a787d8c57b11… · File 483324d3e983919c…

```
 2167 |
 2168 |         self.cv_net = Some(ValueCommitment::derive(spend_value - output_value,
rcv).to_bytes());
 2169 |         Ok(())
```

file_excerpt · zcash_primitives/src/transaction/components/orchard.rs · line 17 · SHA-256 b898fb36dbfe34ab… · File bf0f5a8044330d1c…

```
   16 |     primitives::redpallas::{self, SigType, Signature, SpendAuth, VerificationKey},
   17 |     value::ValueCommitment,
   18 | };
```

file_excerpt · zcash_primitives/src/transaction/components/orchard.rs · line 237 · SHA-256 98f5867a31e27597… · File bf0f5a8044330d1c…

```
  236 |
  237 | pub fn read_value_commitment<R: Read>(mut reader: R) -> io::Result<ValueCommitment> {
  238 |     let mut bytes = [0u8; 32];
```

file_excerpt · zcash_primitives/src/transaction/components/orchard.rs · line 240 · SHA-256 4273fe713482d561… · File bf0f5a8044330d1c…

```
  239 |     reader.read_exact(&mut bytes)?;
  240 |     let cv = ValueCommitment::from_bytes(&bytes);
  241 |
```

file_excerpt · zcash_primitives/src/transaction/components/sapling.rs · line 14 · SHA-256 92e6baa4894cf218… · File ec8e4d0d81abadf1…

```
13 |     note_encryption::{ENC_CIPHERTEXT_SIZE, Zip212Enforcement},
14 |     value::ValueCommitment,
15 | };
```

file_excerpt · zcash_primitives/src/transaction/components/sapling.rs · line 145 · SHA-256 74cbf5fd547cd286… · File ec8e4d0d81abadf1…

```
144 | /// - "Not small order" is enforced here.
145 | fn read_value_commitment<R: Read>(mut reader: R) -> io::Result<ValueCommitment> {
146 |     let mut bytes = [0u8; 32];
```

file_excerpt · zcash_primitives/src/transaction/components/sapling.rs · line 148 · SHA-256 7d95f61598851f8f… · File ec8e4d0d81abadf1…

```
147 |     reader.read_exact(&mut bytes)?;
148 |     let cv = ValueCommitment::from_bytes_not_small_order(&bytes);
149 |
```

file_excerpt · zcash_proofs/src/prover.rs · line 13 · SHA-256 9157c5336593c406… · File a0242a3ed972b317…

```
12 |     prover::{OutputProver, SpendProver},
13 |     value::{NoteValue, ValueCommitTrapdoor},
14 | };
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys and shares must stay valid for how long.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12032 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (15 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

59 reference(s) to ECDSA across 15 file(s), e.g. pczt/src/roles/signer/mod.rs:278. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	pczt/src/roles/signer/mod.rs

Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 18:54 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open pczt/src/roles/signer/mod.rs:278 and inspect the quoted lines; the evidence lists further
files.
```

EVIDENCE

file_excerpt · pczt/src/roles/signer/mod.rs · line 278 · SHA-256 970c986b64283231… · File 2145651f5e27de71…

```
  277 |         index: usize,
  278 |         sk: &secp256k1::SecretKey,
  279 |     ) -> Result<(), Error> {
```

file_excerpt · pczt/src/roles/signer/mod.rs · line 299 · SHA-256 9c9c7a2df266af02… · File 2145651f5e27de71…

```
  298 |         index: usize,
  299 |         signature: secp256k1::ecdsa::Signature,
  300 |     ) -> Result<(), Error> {
```

file_excerpt · zcash_client_backend/src/data_api.rs · line 3935 · SHA-256 96b89b202e7e6ba5… · File c0000e72322a214d…

```
3934 |         _account: <Self as WalletRead>::AccountId,
3935 |         _pubkey: secp256k1::PublicKey,
3936 |     ) -> Result<(), <Self as WalletRead>::Error> {
```

file_excerpt · zcash_client_backend/src/data_api.rs · line 3956 · SHA-256 6ca8dc7f8ad6a965… · File c0000e72322a214d…

```
3955 |         account: <Self as WalletRead>::AccountId,
3956 |         pubkeys: &[secp256k1::PublicKey],
3957 |     ) -> Result<(), <Self as WalletRead>::Error> {
```

file_excerpt · zcash_client_backend/src/data_api/wallet.rs · line 1380 · SHA-256 3b9d26a9135dd6c0… · File ae75d6dfecc19bc4…

```
1379 |     #[cfg(feature = "transparent-key-import")]
1380 |     standalone_transparent_keys: HashMap<TransparentAddress, Vec<secp256k1::SecretKey>>,
1381 | }
```

file_excerpt · zcash_client_backend/src/data_api/wallet.rs · line 1389 · SHA-256 9aa484cc95bda86b… · File ae75d6dfecc19bc4…

```
1388 |             TransparentAddress,
1389 |             Vec<secp256k1::SecretKey>,
1390 |         >,
```

file_excerpt · zcash_client_backend/src/wallet.rs · line 1093 · SHA-256 a2be08c32d9b6f7f… · File 41e038af8afcdd9d…

```
1092 |     pub fn standalone_p2pkh(
1093 |         pubkey: secp256k1::PublicKey,
1094 |         exposure: Exposure,
```

file_excerpt · zcash_client_backend/src/wallet.rs · line 1202 · SHA-256 d1380e8b926885cb... · File 41e038af8afcdd9d...

```
1201 |     #[cfg(feature = "transparent-key-import")]
1202 |     StandalonePubkey(secp256k1::PublicKey),
1203 |
```

file_excerpt · zcash_client_sqlite/src/lib.rs · line 1951 · SHA-256 0e87728a23fc6346... · File c2517b23d072ff31...

```
1950 |         account: <Self as WalletRead>::AccountId,
1951 |         pubkey: secp256k1::PublicKey,
1952 |     ) -> Result<(), <Self as WalletRead>::Error> {
```

file_excerpt · zcash_client_sqlite/src/lib.rs · line 1960 · SHA-256 0df3a5ae0ebfc4ea... · File c2517b23d072ff31...

```
1959 |         account: <Self as WalletRead>::AccountId,
1960 |         pubkeys: &[secp256k1::PublicKey],
1961 |     ) -> Result<(), <Self as WalletRead>::Error> {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12033 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable BLS12-381: transaction and message signing (4 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

5 reference(s) to BLS12-381 across 4 file(s), e.g. zcash_proofs/src/circuit/sprout/mod.rs:340. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:BLS12-381:transaction_signing
Data source	zcash_proofs/src/circuit/sprout/mod.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-04 18:54 UTC

Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open zcash_proofs/src/circuit/sprout/mod.rs:340 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · zcash_proofs/src/circuit/sprout/mod.rs · line 340 · SHA-256 2756f21b7d7a8d3a… · File 2ef74af1024fb6cc…

```
339 |     use bellman::gadgets::test::*;
340 |     use bls12_381::Scalar;
341 |
```

file_excerpt · zcash_proofs/src/lib.rs · line 15 · SHA-256 36b89a654296c810… · File c2050f2e89d86a3c…

```
 14 |
 15 | use bls12_381::Bls12;
 16 | use groth16::{PreparedVerifyingKey, VerifyingKey, prepare_verifying_key};
```

file_excerpt · zcash_proofs/src/prover.rs · line 3 · SHA-256 07a530126847339f… · File a0242a3ed972b317…

```
  2 |
  3 | use bls12_381::Bls12;
  4 | use groth16::Proof;
```

file_excerpt · zcash_proofs/src/prover.rs · line 153 · SHA-256 94cdcb3fa3766000… · File a0242a3ed972b317…

```
152 |         rcv: ValueCommitTrapdoor,
153 |         anchor: bls12_381::Scalar,
154 |         merkle_path: MerklePath,
```

file_excerpt · zcash_proofs/src/sprout.rs · line 4 · SHA-256 927ccca4012b26eb… · File 7b077eeef2f8d854…

```
  3 | use bellman::gadgets::multipack;
  4 | use bls12_381::Bls12;
  5 | use groth16::{Parameters, PreparedVerifyingKey, Proof, create_random_proof};
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-12038 · DEP-KNOWN-ADVISORY

Medium

bytes 1.10.1 is affected by GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007) for crates.io package bytes at version 1.10.1: bytes has integer overflow in BytesMut::reserve. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/bytes@1.10.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-434x-w66g-qw3r
```

EVIDENCE

tool_output · OSV.dev GHSA-434x-w66g-qw3r · manifest · SHA-256 58c8425f679122ea...


```
{"id": "GHSA-434x-w66g-qw3r", "aliases": ["CVE-2026-25541", "RUSTSEC-2026-0007"], "modified":
"2026-09-10T03:50:33.995841351Z", "summary": "bytes has integer overflow in BytesMut::reserve",
"severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/
SI:N/SA:N/E:P"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (1.11.1).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-434x-w66g-qw3r

QSI-2026-12049 · DEP-KNOWN-ADVISORY

Medium

rsa 0.9.6 is affected by GHSA-4grx-2x9w-596c (CVE-2023-49092, GHSA-c38w-74pg-36hr, RUSTSEC-2023-0071)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-4grx-2x9w-596c (CVE-2023-49092, GHSA-c38w-74pg-36hr, RUSTSEC-2023-0071) for crates.io package rsa at version 0.9.6: Marvin Attack: potential key recovery through timing sidechannels. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rsa@0.9.6
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.9 — CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-4grx-2x9w-596c
```

EVIDENCE

tool_output · OSV.dev GHSA-4grx-2x9w-596c · manifest · SHA-256 0ad76c059f8d6aa8...


```
{"id": "GHSA-4grx-2x9w-596c", "aliases": ["CVE-2023-49092", "GHSA-c38w-74pg-36hr",
"RUSTSEC-2023-0071"], "modified": "2026-09-10T03:50:04.002228238Z", "summary": "Marvin Attack:
potential key recovery through timing sidechannels", "severity": [{"type": "CVSS_V3", "score":
"CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:N/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (none published yet; see the advisory for mitigations).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-4grx-2x9w-596c

QSI-2026-12051 · DEP-KNOWN-ADVISORY

Medium

rustls 0.21.12 is affected by GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346) for crates.io package rustls at version 0.21.12: Rustls: Session Debug formatting exposes TLS master secrets and resumption PSKs. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.21.12
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	4.0 — CVSS:3.1/AV:L/AC:H/PR:H/UI:R/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-c8m5-c952-6cm4
```

EVIDENCE

tool_output · OSV.dev GHSA-c8m5-c952-6cm4 · manifest · SHA-256 87554a0cce34470a...


```
{"id": "GHSA-c8m5-c952-6cm4", "aliases": ["RUSTSEC-2026-0346"], "modified":
"2026-10-27T16:15:05.354395696Z", "summary": "Rustls: Session Debug formatting exposes TLS master
secrets and resumption PSKs", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:L/AC:H/PR:H/
UI:R/S:U/C:H/I:N/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-c8m5-c952-6cm4

QSI-2026-12054 · DEP-KNOWN-ADVISORY

Medium

rustls 0.23.25 is affected by GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285) for crates.io package rustls at version 0.23.25: Rustls: TLS 1.3 handshake messages incorrectly accepted across encryption level boundaries. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.23.25
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC

Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	5.3 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-2mjx-qc3c-rqvc
```

EVIDENCE

tool_output · OSV.dev GHSA-2mjx-qc3c-rqvc · manifest · SHA-256 f20e4b21c01d3d75...


```
{"id": "GHSA-2mjx-qc3c-rqvc", "aliases": ["RUSTSEC-2026-0285"], "modified":
"2026-10-27T15:30:05.509305490Z", "summary": "Rustls: TLS 1.3 handshake messages incorrectly
accepted across encryption level boundaries", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/
AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.45).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-2mjx-qc3c-rqvc

QSI-2026-12055 · DEP-KNOWN-ADVISORY

Medium

rustls 0.23.25 is affected by GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346) for crates.io package rustls at version 0.23.25: Rustls: Session Debug formatting exposes TLS master secrets and resumption PSKs. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.23.25
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	4.0 — CVSS:3.1/AV:L/AC:H/PR:H/UI:R/S:U/C:H/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-c8m5-c952-6cm4
```

EVIDENCE

tool_output · OSV.dev GHSA-c8m5-c952-6cm4 · manifest · SHA-256 87554a0cce34470a...


```
{"id": "GHSA-c8m5-c952-6cm4", "aliases": ["RUSTSEC-2026-0346"], "modified":
"2026-10-27T16:15:05.354395696Z", "summary": "Rustls: Session Debug formatting exposes TLS master
secrets and resumption PSKs", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:L/AC:H/PR:H/
UI:R/S:U/C:H/I:N/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-c8m5-c952-6cm4

QSI-2026-12064 · DEP-KNOWN-ADVISORY

Medium

rustls-webpki 0.103.3 is affected by GHSA-pwjx-qhcg-rvj4 (CVE-2026-93602, RUSTSEC-2026-0049)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-pwjx-qhcg-rvj4 (CVE-2026-93602, RUSTSEC-2026-0049) for crates.io package rustls-webpki at version 0.103.3: webpki: CRLs not considered authoritative by Distribution Point due to faulty matching logic. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.103.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	33.1 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	4.4 — CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:H/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-pwjx-qhcg-rvj4
```

EVIDENCE

tool_output · OSV.dev GHSA-pwjx-qhcg-rvj4 · manifest · SHA-256 cfbfa2ab43dcd198...

{"id": "GHSA-pwjx-qhcg-rvj4", "aliases": ["CVE-2026-93602", "RUSTSEC-2026-0049"], "modified": "2026-09-19T03:55:49.529651638Z", "summary": "webpki: CRLs not considered authoritative by Distribution Point due to faulty matching logic", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:H/A:N"}], "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.103.10, 0.104.0-alpha.5).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-pwjx-qhcg-rvj4

QSI-2026-12047 · DEP-KNOWN-ADVISORY

Low

rand 0.8.5 is affected by GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097) for crates.io package rand at version 0.8.5: Rand is unsound with a custom logger using rand::rng(). The audited code names rand::rng: the affected code may be reachable.

Affected asset	dependency:crates.io/rand@0.8.5
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	39.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-cq8v-f236-94qc
```

EVIDENCE

tool_output · OSV.dev GHSA-cq8v-f236-94qc · manifest · SHA-256 a61cde8ef254431f...

{"id": "GHSA-cq8v-f236-94qc", "aliases": ["RUSTSEC-2026-0097"], "modified": "2026-09-10T03:50:43.958966498Z", "summary": "Rand is unsound with a custom logger using rand::rng()", "severity": null, "declared_in_manifest": true}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default

against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.10.1, 0.8.6, 0.9.3).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-cq8v-f236-94qc

QSI-2026-12048 · DEP-KNOWN-ADVISORY

Low

rand 0.9.2 is affected by GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097) for crates.io package rand at version 0.9.2: Rand is unsound with a custom logger using rand::rng(). The audited code names rand::rng: the affected code may be reachable.

Affected asset	dependency:crates.io/rand@0.9.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	39.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-cq8v-f236-94qc
```

EVIDENCE

tool_output · OSV.dev GHSA-cq8v-f236-94qc · manifest · SHA-256 a61cde8ef254431f...


```
{"id": "GHSA-cq8v-f236-94qc", "aliases": ["RUSTSEC-2026-0097"], "modified":
"2026-09-10T03:50:43.958966498Z", "summary": "Rand is unsound with a custom logger using
rand::rng()", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.10.1, 0.8.6, 0.9.3).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-cq8v-f236-94qc

QSI-2026-12050 · DEP-KNOWN-ADVISORY

Low

rsa 0.9.6 is affected by GHSA-9c48-w39g-hm26 (CVE-2026-21895)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-9c48-w39g-hm26 (CVE-2026-21895) for crates.io package rsa at version 0.9.6: rsa crate has potential panic on a prime being equal to 1. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rsa@0.9.6
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-9c48-w39g-hm26
```

EVIDENCE

tool_output · OSV.dev GHSA-9c48-w39g-hm26 · manifest · SHA-256 3e86581a0f0d117f…


```
{"id": "GHSA-9c48-w39g-hm26", "aliases": ["CVE-2026-21895"], "modified":
"2026-09-10T03:50:32.802096391Z", "summary": "rsa crate has potential panic on a prime being equal
to 1", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:L/
SC:N/SI:N/SA:N/E:U"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.9.10).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-9c48-w39g-hm26

QSI-2026-12052 · DEP-KNOWN-ADVISORY

Low

rustls 0.21.12 is affected by GHSA-h2v8-394m-p7vv

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-h2v8-394m-p7vv for crates.io package rustls at version 0.21.12: Rustls: Brotli certificate decompression can allocate an attacker-controlled amount of memory up to 1 GiB before authentication. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.21.12
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	3.7 — CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:N/A:L

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-h2v8-394m-p7vv
```

EVIDENCE

tool_output · OSV.dev GHSA-h2v8-394m-p7vv · manifest · SHA-256 9118726d52fea7ce...


```
{"id": "GHSA-h2v8-394m-p7vv", "aliases": ["RUSTSEC-2026-0348"], "modified":
"2026-10-27T16:15:05.337819489Z", "summary": "Rustls: Brotli certificate decompression can allocate
an attacker-controlled amount of memory up to 1 GiB before authentication.", "severity": [{"type":
"CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:N/A:L"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-h2v8-394m-p7vv

QSI-2026-12053 · DEP-KNOWN-ADVISORY

Low

rustls 0.21.12 is affected by GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349) for crates.io package rustls at version 0.21.12: rustls: An unauthenticated ServerHello poisons the per-name key-exchange record, so later ClientHellos carry only a classical X25519 share. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.21.12
Data source	OSV.dev

Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	3.7 — CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:L/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-w44r-93j7-qv28
```

EVIDENCE

tool_output · OSV.dev GHSA-w44r-93j7-qv28 · manifest · SHA-256 f0fcaa2fa7a56e18…


```
{"id": "GHSA-w44r-93j7-qv28", "aliases": ["RUSTSEC-2026-0349"], "modified":
"2026-10-27T16:15:05.340347056Z", "summary": "rustls: An unauthenticated ServerHello poisons the
per-name key-exchange record, so later ClientHellos carry only a classical X25519 share",
"severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:L/I:N/A:N"}],
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-w44r-93j7-qv28

QSI-2026-12057 · DEP-KNOWN-ADVISORY

Low

rustls 0.23.25 is affected by GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349) for crates.io package rustls at version 0.23.25: rustls: An unauthenticated ServerHello poisons the per-name key-exchange record, so later ClientHellos carry only a classical X25519 share. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls@0.23.25
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	3.7 — CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:L/I:N/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-w44r-93j7-qv28
```

EVIDENCE

tool_output · OSV.dev GHSA-w44r-93j7-qv28 · manifest · SHA-256 f0fcaa2fa7a56e18...


```
{"id": "GHSA-w44r-93j7-qv28", "aliases": ["RUSTSEC-2026-0349"], "modified":
"2026-10-27T16:15:05.340347056Z", "summary": "rustls: An unauthenticated ServerHello poisons the
per-name key-exchange record, so later ClientHellos carry only a classical X25519 share",
"severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:L/I:N/A:N"}],
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-w44r-93j7-qv28

QSI-2026-12059 · DEP-KNOWN-ADVISORY

Low

rustls-webpki 0.101.7 is affected by GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098) for crates.io package rustls-webpki at version 0.101.7: webpki: Name constraints for URI names were incorrectly accepted. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.101.7
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	2.2 — CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-965h-392x-2mh5
```

EVIDENCE

tool_output · OSV.dev GHSA-965h-392x-2mh5 · manifest · SHA-256 7dfbf39859f72638…


```
{"id": "GHSA-965h-392x-2mh5", "aliases": ["CVE-2026-93600", "RUSTSEC-2026-0098"], "modified": "2026-09-19T03:55:32.245575348Z", "summary": "webpki: Name constraints for URI names were incorrectly accepted", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.103.12, 0.104.0-alpha.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-965h-392x-2mh5

QSI-2026-12060 · DEP-KNOWN-ADVISORY

Low

rustls-webpki 0.101.7 is affected by GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099) for crates.io package rustls-webpki at version 0.101.7: webpki: Name constraints were accepted for certificates asserting a wildcard name. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.101.7
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	2.2 — CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-xgp8-3hg3-c2mh
```

EVIDENCE

tool_output · OSV.dev GHSA-xgp8-3hg3-c2mh · manifest · SHA-256 c068ea92e7b545b1…

{"id": "GHSA-xgp8-3hg3-c2mh", "aliases": ["CVE-2026-93601", "RUSTSEC-2026-0099"], "modified": "2026-09-19T03:55:49.146112754Z", "summary": "webpki: Name constraints were accepted for certificates asserting a wildcard name", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N"}], "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.103.12, 0.104.0-alpha.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-xgp8-3hg3-c2mh

QSI-2026-12063 · DEP-KNOWN-ADVISORY

Low

rustls-webpki 0.103.3 is affected by GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098) for crates.io package rustls-webpki at version 0.103.3: webpki: Name constraints for URI names were incorrectly accepted. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.103.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	2.2 — CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-965h-392x-2mh5
```

EVIDENCE

tool_output · OSV.dev GHSA-965h-392x-2mh5 · manifest · SHA-256 7dfbf39859f72638…

{"id": "GHSA-965h-392x-2mh5", "aliases": ["CVE-2026-93600", "RUSTSEC-2026-0098"], "modified": "2026-09-19T03:55:32.245575348Z", "summary": "webpki: Name constraints for URI names were incorrectly accepted", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N"}], "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.103.12, 0.104.0-alpha.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-965h-392x-2mh5

QSI-2026-12065 · DEP-KNOWN-ADVISORY

Low

rustls-webpki 0.103.3 is affected by GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099)

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099) for crates.io package rustls-webpki at version 0.103.3: webpki: Name constraints were accepted for certificates asserting a wildcard name. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.103.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	29.4 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	2.2 — CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-xgp8-3hg3-c2mh
```

EVIDENCE

tool_output · OSV.dev GHSA-xgp8-3hg3-c2mh · manifest · SHA-256 c068ea92e7b545b1...


```
{"id": "GHSA-xgp8-3hg3-c2mh", "aliases": ["CVE-2026-93601", "RUSTSEC-2026-0099"], "modified":
"2026-09-19T03:55:49.146112754Z", "summary": "webpki: Name constraints were accepted for
certificates asserting a wildcard name", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/
AC:H/PR:H/UI:N/S:U/C:N/I:L/A:N"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.103.12, 0.104.0-alpha.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-xgp8-3hg3-c2mh

QSI-2026-12043 · DEP-KNOWN-ADVISORY

Low

keccak 0.1.4 is affected by GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012) for crates.io package keccak at version 0.1.4: Unsoundness in opt-in ARMv8 assembly backend for `keccak`. The affected code sits behind the `asm` Cargo feature, off by default, which no audited manifest enables: the advisory does not apply unless another dependency enables it.

Affected asset	dependency:crates.io/keccak@0.1.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	25.0 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-3288-p39f-rqpv
```

EVIDENCE

tool_output · OSV.dev GHSA-3288-p39f-rqpv · manifest · SHA-256 d5a8f5565101ab9d...


```
{"id": "GHSA-3288-p39f-rqpv", "aliases": ["RUSTSEC-2026-0012"], "modified":
"2026-09-10T03:50:59.332505686Z", "summary": "Unsoundness in opt-in ARMv8 assembly backend for
`keccak`", "severity": [{"type": "CVSS_V4", "score": "CVSS:4.0/AV:L/AC:H/AT:P/PR:N/UI:N/VC:N/VI:L/
VA:N/SC:N/SI:N/SA:N/E:U"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.1.6).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-3288-p39f-rqpv

QSI-2026-12056 · DEP-KNOWN-ADVISORY

Low

rustls 0.23.25 is affected by GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348) for crates.io package rustls at version 0.23.25: Rustls: Brotli certificate decompression can allocate an attacker-controlled amount of memory up to 1 GiB before authentication. The affected code sits behind the `brotli` Cargo feature, off by default, which no audited manifest enables: the advisory does not apply unless another dependency enables it.

Affected asset	dependency:crates.io/rustls@0.23.25
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	25.0 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	3.7 — CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:N/A:L

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-h2v8-394m-p7vv
```

EVIDENCE

tool_output · OSV.dev GHSA-h2v8-394m-p7vv · manifest · SHA-256 9118726d52fea7ce...


```
{"id": "GHSA-h2v8-394m-p7vv", "aliases": ["RUSTSEC-2026-0348"], "modified":
"2026-10-27T16:15:05.337819489Z", "summary": "Rustls: Brotli certificate decompression can allocate
an attacker-controlled amount of memory up to 1 GiB before authentication.", "severity": [{"type":
"CVSS_V3", "score": "CVSS:3.1/AV:N/AC:H/PR:N/UI:N/S:U/C:N/I:N/A:L"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.23.46, 0.24.0-dev.2).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-h2v8-394m-p7vv

QSI-2026-12036 · DEP-ADVISORY-UNRATED

Low

anyhow 1.0.100 is affected by RUSTSEC-2026-0190 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0190 for crates.io package anyhow at version 1.0.100: Unsoundness in `Error::downcast_mut()`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (anyhow::Error::downcast_mut); anyhow comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/anyhow@1.0.100
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0190
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0190 · manifest · SHA-256 12d4b3eeb5a7dc67…


```
{"id": "RUSTSEC-2026-0190", "aliases": [], "modified": "2026-06-29T14:15:03.650725453Z", "summary": "Unsoundness in `Error::downcast_mut()`", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (1.0.103) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0190

QSI-2026-12039 · DEP-ADVISORY-UNRATED

Low

crossbeam-epoch 0.9.18 is affected by RUSTSEC-2026-0204 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0204 for crates.io package crossbeam-epoch at version 0.9.18: Invalid pointer dereference in `fmt::Pointer` impl for `Atomic` and `Shared` when the underlying pointer is invalid. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/crossbeam-epoch@0.9.18
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0204
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0204 · manifest · SHA-256 87f069b4fa857d22...


```
{"id": "RUSTSEC-2026-0204", "aliases": [], "modified": "2026-07-06T22:30:04.159269138Z", "summary":
"Invalid pointer dereference in `fmt::Pointer` impl for `Atomic` and `Shared` when the underlying
pointer is invalid", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.9.20) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0204

QSI-2026-12040 · DEP-ADVISORY-UNRATED

Low

crypto-bigint 0.5.5 is affected by RUSTSEC-2026-0387 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0387 for crates.io package crypto-bigint at version 0.5.5: Multiple timing variability issues in operations intended to be constant-time. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/crypto-bigint@0.5.5

Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0387
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0387 · manifest · SHA-256 eab63427da988e95...


```
{"id": "RUSTSEC-2026-0387", "aliases": [], "modified": "2026-10-24T03:45:02.365137418Z", "summary":
"Multiple timing variability issues in operations intended to be constant-time", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.7.6) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0387

QSI-2026-12041 · DEP-ADVISORY-UNRATED

Low

ctutils 0.4.2 is affected by RUSTSEC-2026-0385 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0385 for crates.io package ctutils at version 0.4.2: Non-constant-time mask generation and `u128` comparisons. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/ctutils@0.4.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0385
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0385 · manifest · SHA-256 200470326c8a5ad1...


```
{"id": "RUSTSEC-2026-0385", "aliases": [], "modified": "2026-10-23T08:30:02.389575143Z", "summary":
"Non-constant-time mask generation and `u128` comparisons", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.4.4) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0385

QSI-2026-12042 · DEP-ADVISORY-UNRATED

Low

event-listener 5.3.1 is affected by RUSTSEC-2026-0221 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0221 for crates.io package event-listener at version 5.3.1: `event-listener` allows `!Send` tags to cross thread boundaries via `StackSlot`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/event-listener@5.3.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0221
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0221 · manifest · SHA-256 984b4e4e87b4e2cc...


```
{"id": "RUSTSEC-2026-0221", "aliases": [], "modified": "2026-07-31T12:15:04.629484145Z", "summary":
"`event-listener` allows `!Send` tags to cross thread boundaries via `StackSlot`", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (5.4.2) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0221

QSI-2026-12044 · DEP-ADVISORY-UNRATED

Low

memmap2 0.9.4 is affected by RUSTSEC-2026-0186 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0186 for crates.io package memmap2 at version 0.9.4: Unchecked pointer offset in crate `memmap2`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (memmap2::Mmap::advise_range, memmap2::Mmap::unchecked_advise_range, memmap2::MmapMut::advise_range, memmap2::MmapMut::flush_async_range, memmap2::MmapMut::flush_range, memmap2::MmapMut::unchecked_advise_range); memmap2 comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/memmap2@0.9.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0186
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0186 · manifest · SHA-256 70b02eedb4e58b99...


```
{"id": "RUSTSEC-2026-0186", "aliases": [], "modified": "2026-06-22T18:15:03.600280397Z", "summary":
"Unchecked pointer offset in crate `memmap2`", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.9.11) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0186

QSI-2026-12061 · DEP-ADVISORY-UNRATED

Low

rustls-webpki 0.101.7 is affected by RUSTSEC-2026-0382 (CVE-2026-111943) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0382 (CVE-2026-111943) for crates.io package rustls-webpki at version 0.101.7: Certificate verification may accept distrusted intermediates. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.101.7
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0382
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0382 · manifest · SHA-256 4c972eb1d77abf0d...

{"id": "RUSTSEC-2026-0382", "aliases": ["BIT-golang-2026-111943", "CVE-2026-111943", "GO-2026-6809", "RUSTSEC-2026-0381", "RUSTSEC-2026-0383"], "modified": "2026-10-29T10:55:54.096394276Z", "summary": "Certificate verification may accept distrusted intermediates", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0382

QSI-2026-12066 · DEP-ADVISORY-UNRATED

Low

rustls-webpki 0.103.3 is affected by RUSTSEC-2026-0382 (CVE-2026-111943) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0382 (CVE-2026-111943) for crates.io package rustls-webpki at version 0.103.3: Certificate verification may accept distrusted intermediates. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/rustls-webpki@0.103.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0382
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0382 · manifest · SHA-256 4c972eb1d77abf0d...

{"id": "RUSTSEC-2026-0382", "aliases": ["BIT-golang-2026-111943", "CVE-2026-111943", "GO-2026-6809", "RUSTSEC-2026-0381", "RUSTSEC-2026-0383"], "modified": "2026-10-29T10:55:54.096394276Z", "summary": "Certificate verification may accept distrusted intermediates", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0382

QSI-2026-12069 · DEP-ADVISORY-UNRATED

Low

webpki-roots 0.25.4 is affected by RUSTSEC-2026-0381 (CVE-2026-111943) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0381 (CVE-2026-111943) for crates.io package webpki-roots at version 0.25.4: Bundled root certificates allow trust in distrusted emSign intermediates. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/webpki-roots@0.25.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0381
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0381 · manifest · SHA-256 34a59ae82e2e8e2c...


```
{"id": "RUSTSEC-2026-0381", "aliases": ["BIT-golang-2026-111943", "CVE-2026-111943", "GO-2026-6809",
"RUSTSEC-2026-0382", "RUSTSEC-2026-0383"], "modified": "2026-10-29T10:55:54.096394276Z", "summary":
"Bundled root certificates allow trust in distrusted emSign intermediates", "severity": null,
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.26.11, 1.0.10) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0381

QSI-2026-12070 · DEP-ADVISORY-UNRATED

Low

webpki-roots 1.0.3 is affected by RUSTSEC-2026-0381 (CVE-2026-111943) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0381 (CVE-2026-111943) for crates.io package webpki-roots at version 1.0.3: Bundled root certificates allow trust in distrusted emSign intermediates. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/webpki-roots@1.0.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0381
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0381 · manifest · SHA-256 34a59ae82e2e8e2c...


```
{"id": "RUSTSEC-2026-0381", "aliases": ["BIT-golang-2026-111943", "CVE-2026-111943", "GO-2026-6809",
"RUSTSEC-2026-0382", "RUSTSEC-2026-0383"], "modified": "2026-10-29T10:55:54.096394276Z", "summary":
"Bundled root certificates allow trust in distrusted emSign intermediates", "severity": null,
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.26.11, 1.0.10) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0381

QSI-2026-12071 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0395 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0395 for crates.io package zstd-safe at version 7.1.0: Use-after-free with `CParameter::StableInBuffer`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::CCtx::set_parameter); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0395
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0395 · manifest · SHA-256 728850176f43ecf2...


```
{"id": "RUSTSEC-2026-0395", "aliases": [], "modified": "2026-11-02T09:30:03.420559993Z", "summary":
"Use-after-free with `CParameter::StableInBuffer`", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0395

QSI-2026-12072 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0397 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0397 for crates.io package zstd-safe at version 7.1.0: `OutBuffer::around_pos` exposes uninitialized memory. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::OutBuffer::around_pos); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0397
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0397 · manifest · SHA-256 4ad3bc187fa64e24...


```
{"id": "RUSTSEC-2026-0397", "aliases": [], "modified": "2026-11-02T09:30:03.445177143Z", "summary": "`OutBuffer::around_pos` exposes uninitialized memory", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0397

QSI-2026-12073 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0399 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0399 for crates.io package zstd-safe at version 7.1.0: Use-after-free through `init_using_cdict` and `init_using_ddict`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::CCtx::init_using_cdict, zstd_safe::DCtx::init_using_ddict); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0399
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0399 · manifest · SHA-256 58cd19f1b734988f...


```
{"id": "RUSTSEC-2026-0399", "aliases": [], "modified": "2026-11-02T09:30:03.328955732Z", "summary":
"Use-after-free through `init_using_cdict` and `init_using_ddict`", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0399

QSI-2026-12074 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0400 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0400 for crates.io package zstd-safe at version 7.1.0: Use-after-free of a dictionary after a one-shot call on a context with an open stream. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::DCtx::decompress,

zstd_safe::DCtx::decompress_using_ddict, zstd_safe::DCtx::decompress_using_dict); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0400
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0400 · manifest · SHA-256 00f83f7c4931137d...


```
{"id": "RUSTSEC-2026-0400", "aliases": [], "modified": "2026-11-02T09:30:03.428341570Z", "summary": "Use-after-free of a dictionary after a one-shot call on a context with an open stream", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0400

QSI-2026-12075 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0401 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0401 for crates.io package zstd-safe at version 7.1.0: Out-of-bounds read in `train_from_buffer` when sample sizes overflow. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::train_from_buffer); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0401
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0401 · manifest · SHA-256 dd6e9bd27e75cf4c...


```
{"id": "RUSTSEC-2026-0401", "aliases": [], "modified": "2026-11-02T09:30:03.353698025Z", "summary":
"Out-of-bounds read in `train_from_buffer` when sample sizes overflow", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0401

QSI-2026-12076 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0402 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0402 for crates.io package zstd-safe at version 7.1.0: Heap buffer overflow in `CCtx::try_clone` on a multithreaded context. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::CCtx::try_clone); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0402
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0402 · manifest · SHA-256 464c8c61aa6d0a29...


```
{"id": "RUSTSEC-2026-0402", "aliases": [], "modified": "2026-11-02T09:30:04.162155944Z", "summary":
"Heap buffer overflow in `CCtx::try_clone` on a multithreaded context", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0402

QSI-2026-12077 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0403 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0403 for crates.io package zstd-safe at version 7.1.0: Uninitialized memory exposed when streaming decompression of a legacy frame fails. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::DCtx::decompress_stream); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0403
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0403 · manifest · SHA-256 09699f0d23ac15a1...


```
{"id": "RUSTSEC-2026-0403", "aliases": [], "modified": "2026-11-02T09:30:04.136442771Z", "summary":
"Uninitialized memory exposed when streaming decompression of a legacy frame fails", "severity":
null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0403

QSI-2026-12078 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0404 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0404 for crates.io package zstd-safe at version 7.1.0: Double free in `DCtx::try_clone`. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::DCtx::try_clone); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0404
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0404 · manifest · SHA-256 e265647b9f2c582e...

{"id": "RUSTSEC-2026-0404", "aliases": [], "modified": "2026-11-02T09:30:04.168804805Z", "summary": "Double free in `DCtx::try_clone`", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0404

QSI-2026-12079 · DEP-ADVISORY-UNRATED

Low

zstd-safe 7.1.0 is affected by RUSTSEC-2026-0406 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0406 for crates.io package zstd-safe at version 7.1.0: Use-after-free when changing a multithreaded context with an unfinished frame. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The audited code does not name the affected functions (zstd_safe::CCtx::disable_dictionary, zstd_safe::CCtx::init, zstd_safe::CCtx::load_dictionary, zstd_safe::CCtx::ref_cdict, zstd_safe::CCtx::ref_prefix, zstd_safe::CCtx::reset …); zstd-safe comes in through other dependencies, which may still call them.

Affected asset	dependency:crates.io/zstd-safe@7.1.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0406
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0406 · manifest · SHA-256 f08bb0f8849019aa…

{"id": "RUSTSEC-2026-0406", "aliases": [], "modified": "2026-11-02T09:30:03.591194710Z", "summary": "Use-after-free when changing a multithreaded context with an unfinished frame", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (7.4.0, 8.2.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0406

QSI-2026-12080 · DEP-ADVISORY-UNRATED

Low

zstd-sys 2.0.10+zstd.1.5.6 is affected by RUSTSEC-2026-0410 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0410 for crates.io package zstd-sys at version 2.0.10+zstd.1.5.6: WebAssembly allocator shim corrupts memory on allocation failure. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined. It affects only wasm32 targets.

Affected asset	dependency:crates.io/zstd-sys@2.0.10+zstd.1.5.6
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	20.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0410
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0410 · manifest · SHA-256 1e892df4ce7c9911…


```
{"id": "RUSTSEC-2026-0410", "aliases": [], "modified": "2026-10-26T14:00:05.788985783Z", "summary":
"WebAssembly allocator shim corrupts memory on allocation failure", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (2.1.2) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0410

QSI-2026-12068 · DEP-ADVISORY-UNRATED

Info

tokio 1.50.0 is affected by RUSTSEC-2026-0386 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0386 for crates.io package tokio at version 1.50.0: `AsyncFd::new` and related constructors violate I/O safety. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (tokio::io::unix::AsyncFd::new, tokio::io::unix::AsyncFd::try_new, tokio::io::unix::AsyncFd::try_with_interest, tokio::io::unix::AsyncFd::with_interest) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:crates.io/tokio@1.50.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0386
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0386 · manifest · SHA-256 1ebb594fd9923dc0...


```
{"id": "RUSTSEC-2026-0386", "aliases": [], "modified": "2026-10-23T14:45:03.847864944Z", "summary":
"`AsyncFd::new` and related constructors violate I/O safety", "severity": null,
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (1.51.6, 1.53.3) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0386

QSI-2026-12081 · DEP-ADVISORY-DEV

Info

3 advisories in 2 development or build-only crates.io package(s)

Improvement recommendation | Confidence: Medium | preliminary — requires further verification

OSV.dev lists 3 advisories (2 rated high or critical) for 2 crates.io package(s) that only development or build tooling pulls in (Cargo.lock): e.g. gumdrop@0.8.1 [dev]: RUSTSEC-2026-0214; quick-xml@0.26.0 [dev]: RUSTSEC-2026-0194, RUSTSEC-2026-0195. The manifests mark them as such (devDependencies, "dev": true in package-lock.json, pnpm importers sections, [dev-dependencies]/[build-dependencies] reached through Cargo.lock, or — for yarn.lock, which records no scope at all — package.json reached through the lock graph), so they are not part of what ships.

Affected asset	dependencies:crates.io:non-runtime
Data source	OSV.dev
Detection method	OSV.dev querybatch lookup of exact versions; scope from devDependencies, package-lock.json dev flags, pnpm importers sections, the Cargo.lock graph rooted at the Cargo.toml tables, and — for yarn.lock, which records no scope — the lock graph walked from package.json. Where a runtime requirement cannot be resolved in a lock, that lock is left unclassified rather than risking a package being set aside wrongly.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	15.5 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

They do not reach users of the product. They can affect developer machines and CI (supply chain): a compromised or vulnerable tool can tamper with builds.

REPRODUCTION STEPS

```
Look the packages listed in the evidence up at https://osv.dev and check their scope in Cargo.lock.
```

EVIDENCE

tool_output · OSV.dev (crates.io) · Cargo.lock · SHA-256 616397a43959bf6a...

```
gumdrop@0.8.1 [dev]: RUSTSEC-2026-0214
quick-xml@0.26.0 [dev]: RUSTSEC-2026-0194, RUSTSEC-2026-0195
```

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Scopes come from the manifests; a registry crate's own build dependencies cannot be told apart in Cargo.lock, so the runtime set may be larger than reality, never smaller.

RECOMMENDATION

Update the tooling during routine maintenance; prioritise packages that run in CI with secrets or that produce release artefacts.

RETEST PROCEDURE

Re-run the dependency scan after updating.

OSV.dev — open-source vulnerability database and API

QSI-2026-12034 · DEP-NO-PQC-LIBRARY

Info

No post-quantum capable cryptographic library detected

Insufficient data | Confidence: Medium | preliminary — requires further verification

Across 21 dependency manifests, QSI found crypto libraries (bellman, bls12_381, curve25519-dalek, ecdsa, ed25519-dalek, halo2_gadgets, halo2_poseidon, halo2_proofs, jubjub, orchard, p256, p384, pasta_curves, reddsa, redjubjub) but none from its PQC-capable list.

Affected asset	dependencies
Data source	dependency manifests
Detection method	Dependency name matching.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	13.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Readiness indicator only: PQC support may come from the platform (OpenSSL, OS, cloud KMS) rather than application dependencies.

REPRODUCTION STEPS

```
Review the dependency manifests listed in the evidence.
```

EVIDENCE

dependency · Cargo.lock, Cargo.toml, components/eip681/Cargo.toml, components/equihash/Cargo.toml, components/f4jumble/Cargo.toml, components/payment_uri/Cargo.toml, components/zcash_address/Cargo.toml, components/zcash_encoding/Cargo.toml, components/zcash_protocol/Cargo.toml, components/zip321/Cargo.toml, pczt/Cargo.toml, zcash/Cargo.toml, zcash_client_backend/Cargo.toml, zcash_client_sqlite/Cargo.toml, zcash_history/Cargo.toml, zcash_keys/Cargo.toml, zcash_pool_migration/Cargo.toml, zcash_pool_migration_m · SHA-256 f4dd81bc784f3fbe…

```
Crypto libraries: bellman, bls12_381, curve25519-dalek, ecdsa, ed25519-dalek, halo2_gadgets,
halo2_poseidon, halo2_proofs, jubjub, orchard, p256, p384, pasta_curves, reddsa, redjubjub, ring,
rsa, rustls, sapling-crypto, secp256k1, sinsemilla, x25519-dalek, zcash_client_backend, zcash_keys,
zcash_note_encryption, zcash_primitives, zcash_proofs, zcash_protocol, zcash_spec, zip32
```

LIMITATIONS OF THE CHECK

Absence of evidence is not evidence of absence: system libraries and transitive dependencies may not be listed.

RECOMMENDATION

Confirm where PQC support will come from (TLS terminator, KMS/HSM, library upgrade).

RETEST PROCEDURE

Re-scan after upgrades.

FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM); FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)

QSI-2026-12035 · DEP-ADVISORY-INFORMATIONAL

Info

adler 1.0.2: informational advisory RUSTSEC-2025-0056 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2025-0056 for crates.io package adler at version 1.0.2 as informational (unmaintained): adler crate is unmaintained, use adler2 instead. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/adler@1.0.2
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2025-0056
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2025-0056 · manifest · SHA-256 e9b7ae0431d69f62...


```
{"id": "RUSTSEC-2025-0056", "aliases": [], "modified": "2025-09-05T09:05:48Z", "summary": "adler
crate is unmaintained, use adler2 instead", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2025-0056

QSI-2026-12037 · DEP-ADVISORY-INFORMATIONAL

Info

bincode 2.0.1: informational advisory RUSTSEC-2025-0141 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2025-0141 for crates.io package bincode at version 2.0.1 as informational (unmaintained): Bincode is unmaintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/bincode@2.0.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2025-0141
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2025-0141 · manifest · SHA-256 532525d6fd3f2cb4...


```
{"id": "RUSTSEC-2025-0141", "aliases": [], "modified": "2026-01-16T11:07:28.867553Z", "summary":
"Bincode is unmaintained", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2025-0141

QSI-2026-12045 · DEP-ADVISORY-INFORMATIONAL

Info

paste 1.0.15: informational advisory RUSTSEC-2024-0436 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2024-0436 for crates.io package paste at version 1.0.15 as informational (unmaintained): paste - no longer maintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/paste@1.0.15
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2024-0436
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2024-0436 · manifest · SHA-256 ea867adb8ba26c96...

{"id": "RUSTSEC-2024-0436", "aliases": [], "modified": "2026-01-27T22:03:15.694766Z", "summary": "paste - no longer maintained", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2024-0436

QSI-2026-12046 · DEP-ADVISORY-INFORMATIONAL

Info

proc-macro-error2 2.0.1: informational advisory RUSTSEC-2026-0173 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2026-0173 for crates.io package proc-macro-error2 at version 2.0.1 as informational (unmaintained): proc-macro-error2 is unmaintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/proc-macro-error2@2.0.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-04 18:54 UTC
Priority score	11.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0173
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0173 · manifest · SHA-256 0ecc182141bd9eff...

{"id": "RUSTSEC-2026-0173", "aliases": [], "modified": "2026-06-07T21:00:04.318567692Z", "summary": "proc-macro-error2 is unmaintained", "severity": null, "declared_in_manifest": false}

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2026-0173

Evidence

ID	Type	Data source	SHA-256
QSI-2026-12058	tool_output	OSV.dev GHSA-82j2-j2ch-gfr8 (manifest)	a2f685e85eb5c808cb9d2de44620be02b2f87b43da592ad8769d4abeedd
QSI-2026-12062	tool_output	OSV.dev GHSA-82j2-j2ch-gfr8 (manifest)	a2f685e85eb5c808cb9d2de44620be02b2f87b43da592ad8769d4abeedd
QSI-2026-12067	tool_output	OSV.dev GHSA-7gcf-g7xr-8hxj (manifest)	b42c751004bfb387a1f24d4e9b23157ae70dc8f33fdb503f87f6d042550
QSI-2026-12031	file_excerpt	pczt/src/orchard.rs (line 23)	ceab0b6e960329ba36b083a225009b5f573bb5a46353fd5bc02ebee6c19
QSI-2026-12031	file_excerpt	pczt/src/orchard.rs (line 2161)	6b6748f70bf576cd97e6461d8a6c1dfc5e0cd8335a7a03c4379d504ff32
QSI-2026-12031	file_excerpt	pczt/src/orchard.rs (line 2168)	14e2a787d8c57b11ecb388d872ab15526a24526cbad0174a72c6e99de40
QSI-2026-12031	file_excerpt	zcash_primitives/src/transaction/components/orchard.rs (line 17)	b898fb36dbfe34ab7cb1b164bb079efb5379f9fbfa5c96a4e7ec5553008
QSI-2026-12031	file_excerpt	zcash_primitives/src/transaction/components/orchard.rs (line 237)	98f5867a31e27597c3c01fb15d4bff203c7c50e97aeb84452538a660818
QSI-2026-12031	file_excerpt	zcash_primitives/src/transaction/components/orchard.rs (line 240)	4273fe713482d561f58f7ead2f139ac70d03ea9dbad8e017bd7461264af
QSI-2026-12031	file_excerpt	zcash_primitives/src/transaction/components/sapling.rs (line 14)	92e6baa4894cf218ffaee5a99ea7a496207f7809ceb7dd93bdad9617eac
QSI-2026-12031	file_excerpt	zcash_primitives/src/transaction/components/sapling.rs (line 145)	74cbf5fd547cd286f9ab13c63b7ff0d2a3e65edce93bf842d9941a5de53
QSI-2026-12031	file_excerpt	zcash_primitives/src/transaction/components/sapling.rs (line 148)	7d95f61598851f8f97630b7d4273b0d979553c0dd4163a84350ec07504f
QSI-2026-12031	file_excerpt	zcash_proofs/src/prover.rs (line 13)	9157c5336593c406d960a2e8c63876e8949011faf52f1090e1a110015c7
QSI-2026-12032	file_excerpt	pczt/src/roles/signer/mod.rs (line 278)	970c986b642832312ead47151a60f06858a4b568e9da6447791c151d9c3

QSI-2026-12032	file_excerpt	pczt/src/roles/signer/mod.rs (line 299)	9c9c7a2df266af024bfa9215a017640135b9c1b6f6054baeca62134856b
QSI-2026-12032	file_excerpt	zcash_client_backend/src/data_api.rs (line 3935)	96b89b202e7e6ba5fb6a218ecdfd6e64c4c7f004c23a197ce1a7c0e358f
QSI-2026-12032	file_excerpt	zcash_client_backend/src/data_api.rs (line 3956)	6ca8dc7f8ad6a96576c26f6abb806e86471829e7355b33eddfdef8e5002
QSI-2026-12032	file_excerpt	zcash_client_backend/src/data_api/wallet.rs (line 1380)	3b9d26a9135dd6c0de7940c206e5fe0f7700515582a4903f60cef7b3fbe
QSI-2026-12032	file_excerpt	zcash_client_backend/src/data_api/wallet.rs (line 1389)	9aa484cc95bda86ba52610e3ef95bd1009860a3b691a37e0f12d6aae37e
QSI-2026-12032	file_excerpt	zcash_client_backend/src/wallet.rs (line 1093)	a2be08c32d9b6f7fab6974e5d502963b486a999af71b05061fca62ffb95
QSI-2026-12032	file_excerpt	zcash_client_backend/src/wallet.rs (line 1202)	d1380e8b926885cbf4d779915616b5696ec37a900f98f0a1505563ba0c2
QSI-2026-12032	file_excerpt	zcash_client_sqlite/src/lib.rs (line 1951)	0e87728a23fc63465b4ae99448d66aec1aee00fc60e55775df2fb66f029
QSI-2026-12032	file_excerpt	zcash_client_sqlite/src/lib.rs (line 1960)	0df3a5ae0ebfc4ea62b057c34c50c3fc68a6a763fcd7461a55656ffe68e
QSI-2026-12033	file_excerpt	zcash_proofs/src/circuit/sprout/mod.rs (line 340)	2756f21b7d7a8d3a022baf241563ed66e885559fd1f4c8cf87ae87c061b
QSI-2026-12033	file_excerpt	zcash_proofs/src/lib.rs (line 15)	36b89a654296c8100d863d399d7c6fe1c0fda438afc6aea4b7621d0eeee
QSI-2026-12033	file_excerpt	zcash_proofs/src/prover.rs (line 3)	07a530126847339f44470f24e6e9fc74c8ca1af51c614ea77c466f5bc8c
QSI-2026-12033	file_excerpt	zcash_proofs/src/prover.rs (line 153)	94cdcb3fa3766000a515cf37e8aa3a5d2a1c0cd7a54b33a3604a8255764
QSI-2026-12033	file_excerpt	zcash_proofs/src/sprout.rs (line 4)	927ccca4012b26eb057e2bbe1d966d250cb67af186853d25e3efbf9bbd7
QSI-2026-12038	tool_output	OSV.dev GHSA-434x-w66g-qw3r (manifest)	58c8425f679122ea78208908d7c42d7c0f6eb030e5fbaef6d65097f3e67
QSI-2026-12049	tool_output	OSV.dev GHSA-4grx-2x9w-596c (manifest)	0ad76c059f8d6aa8bafc0408cce98e5ddfd2906fe7f762bb486112aefdf
QSI-2026-12051	tool_output	OSV.dev GHSA-c8m5-c952-6cm4 (manifest)	87554a0cce34470a221f04ad57566b2b512ff6299ad3ee95b190df659d6
QSI-2026-12054	tool_output	OSV.dev GHSA-2mjx-qc3c-rqvc (manifest)	f20e4b21c01d3d75460ac8d90a54d541a1a34187a4f05b501e4f46c27c5

QSI-2026-12055	tool_output	OSV.dev GHSA-c8m5-c952-6cm4 (manifest)	87554a0cce34470a221f04ad57566b2b512ff6299ad3ee95b190df659d6
QSI-2026-12064	tool_output	OSV.dev GHSA-pwjx-qhcg-rvj4 (manifest)	cfbfa2ab43dcd198e7076fcab3a8ad64c1c5d529fc6135a1b8e5d707cb9
QSI-2026-12047	tool_output	OSV.dev GHSA-cq8v-f236-94qc (manifest)	a61cde8ef254431f750cff43ce8ebe999ba61525990761c3c15b83d24b6
QSI-2026-12048	tool_output	OSV.dev GHSA-cq8v-f236-94qc (manifest)	a61cde8ef254431f750cff43ce8ebe999ba61525990761c3c15b83d24b6
QSI-2026-12050	tool_output	OSV.dev GHSA-9c48-w39g-hm26 (manifest)	3e86581a0f0d117fc19d0a64e364348f29c6c9ecc2309c5c504009aaf90
QSI-2026-12052	tool_output	OSV.dev GHSA-h2v8-394m-p7vv (manifest)	9118726d52fea7cee09a6b4a21a10d2a7bab860f9762fff4658cb78db04
QSI-2026-12053	tool_output	OSV.dev GHSA-w44r-93j7-qv28 (manifest)	f0fcaa2fa7a56e188458f39e55b23e999d0c7da6d3c1294c014c64ab37b
QSI-2026-12057	tool_output	OSV.dev GHSA-w44r-93j7-qv28 (manifest)	f0fcaa2fa7a56e188458f39e55b23e999d0c7da6d3c1294c014c64ab37b
QSI-2026-12059	tool_output	OSV.dev GHSA-965h-392x-2mh5 (manifest)	7dfbf39859f726382fb5bacff8569cd96b6823e01f426668833099ff6d7
QSI-2026-12060	tool_output	OSV.dev GHSA-xgp8-3hg3-c2mh (manifest)	c068ea92e7b545b187eef89198d9b7ef2f675512c425f0664366565735f
QSI-2026-12063	tool_output	OSV.dev GHSA-965h-392x-2mh5 (manifest)	7dfbf39859f726382fb5bacff8569cd96b6823e01f426668833099ff6d7
QSI-2026-12065	tool_output	OSV.dev GHSA-xgp8-3hg3-c2mh (manifest)	c068ea92e7b545b187eef89198d9b7ef2f675512c425f0664366565735f
QSI-2026-12043	tool_output	OSV.dev GHSA-3288-p39f-rqpv (manifest)	d5a8f5565101ab9d60521142f31c54583db5b87e9a59fc99e1fa93d6c2e
QSI-2026-12056	tool_output	OSV.dev GHSA-h2v8-394m-p7vv (manifest)	9118726d52fea7cee09a6b4a21a10d2a7bab860f9762fff4658cb78db04
QSI-2026-12036	tool_output	OSV.dev RUSTSEC-2026-0190 (manifest)	12d4b3eeb5a7dc67549bc112051813542621c4c3bb2d3ddfb353886e687
QSI-2026-12039	tool_output	OSV.dev RUSTSEC-2026-0204 (manifest)	87f069b4fa857d22ba2571deff7f541f4622924fb5fc8545cd46d35b7d4

QSI-2026-12040	tool_output	OSV.dev RUSTSEC-2026-0387 (manifest)	eab63427da988e95becf061fd8169abcb9634f743aeeb7567dfa2ba2667
QSI-2026-12041	tool_output	OSV.dev RUSTSEC-2026-0385 (manifest)	200470326c8a5ad19deb3639dff41d33749b1153097b6b87ef95e4beec7
QSI-2026-12042	tool_output	OSV.dev RUSTSEC-2026-0221 (manifest)	984b4e4e87b4e2cc5992a06daf6b6466ee08279d927a38dca105b75fa85
QSI-2026-12044	tool_output	OSV.dev RUSTSEC-2026-0186 (manifest)	70b02eedb4e58b99412d14613c1a9dc60f7cb08fd4835cd298286af4dc4
QSI-2026-12061	tool_output	OSV.dev RUSTSEC-2026-0382 (manifest)	4c972eb1d77abf0d1c39f2b402c5fea32a5dacf7dbbeac638e7549ce780
QSI-2026-12066	tool_output	OSV.dev RUSTSEC-2026-0382 (manifest)	4c972eb1d77abf0d1c39f2b402c5fea32a5dacf7dbbeac638e7549ce780
QSI-2026-12069	tool_output	OSV.dev RUSTSEC-2026-0381 (manifest)	34a59ae82e2e8e2cde2fe94cd0e4c1ee7c6bd3656c3229fa2fa49bb4c3c
QSI-2026-12070	tool_output	OSV.dev RUSTSEC-2026-0381 (manifest)	34a59ae82e2e8e2cde2fe94cd0e4c1ee7c6bd3656c3229fa2fa49bb4c3c
QSI-2026-12071	tool_output	OSV.dev RUSTSEC-2026-0395 (manifest)	728850176f43ecf2b9c562524c791d0da3abc31547980bb333caeadf51e
QSI-2026-12072	tool_output	OSV.dev RUSTSEC-2026-0397 (manifest)	4ad3bc187fa64e24d070e53c5d01373b2aba040ea27757bbf26f8efea74
QSI-2026-12073	tool_output	OSV.dev RUSTSEC-2026-0399 (manifest)	58cd19f1b734988f6219b3659cd613c349a5c6693e08cb4e8b9846d1f92
QSI-2026-12074	tool_output	OSV.dev RUSTSEC-2026-0400 (manifest)	00f83f7c4931137d6b18e25256c039f2c52bd627946f5f05f74c2cc7b26
QSI-2026-12075	tool_output	OSV.dev RUSTSEC-2026-0401 (manifest)	dd6e9bd27e75cf4c8d9a5333bbc4dda81ef2f2990271b711f964d6bc829
QSI-2026-12076	tool_output	OSV.dev RUSTSEC-2026-0402 (manifest)	464c8c61aa6d0a29b646ba665145c6d256c18786ece14e1921252cecf4a
QSI-2026-12077	tool_output	OSV.dev RUSTSEC-2026-0403 (manifest)	09699f0d23ac15a18937ce02a1b778a246ec661449852b62d6415d8d60e

QSI-2026-12078	tool_output	OSV.dev RUSTSEC-2026-0404 (manifest)	e265647b9f2c582eebfcac7889bee04fada79ea833147205829d693f4c6
QSI-2026-12079	tool_output	OSV.dev RUSTSEC-2026-0406 (manifest)	f08bb0f8849019aa21c995ae08e9c10ff924c83b0464e87d3143ab07935
QSI-2026-12080	tool_output	OSV.dev RUSTSEC-2026-0410 (manifest)	1e892df4ce7c9911f01547e4a80d1fb3ea5a27b8c1c89c0514917e08bbf
QSI-2026-12068	tool_output	OSV.dev RUSTSEC-2026-0386 (manifest)	1ebb594fd9923dc03209b41738d3f9cf8f324941e8844709473c643cd24
QSI-2026-12081	tool_output	OSV.dev (crates.io) (Cargo.lock)	616397a43959bf6a4ed1107e089d8fabf0b48f7c38a24a6c6d335201523
QSI-2026-12034	dependency	Cargo.lock, Cargo.toml, components/eip681/ Cargo.toml, components/equihash/ Cargo.toml, components/f4jumble/ Cargo.toml, components/ payment_uri/Cargo.toml, components/ zcash_address/ Cargo.toml, components/ zcash_encoding/ Cargo.toml, components/ zcash_protocol/ Cargo.toml, components/zip321/ Cargo.toml, pczt/ Cargo.toml, zcash/ Cargo.toml, zcash_client_backend/ Cargo.toml, zcash_client_sqlite/ Cargo.toml, zcash_history/ Cargo.toml, zcash_keys/ Cargo.toml, zcash_pool_migration/ Cargo.toml, zcash_pool_migration_m	f4dd81bc784f3fbe0d0634487bd1a57a8b50abbe8a53b57e1ac025819a3
QSI-2026-12035	tool_output	OSV.dev RUSTSEC-2025-0056 (manifest)	e9b7ae0431d69f626bc876560ccaff44bb57b35757fd9296256424ca73e

QSI-2026-12037	tool_output	OSV.dev RUSTSEC-2025-0141 (manifest)	532525d6fd3f2cb4701cae8133e80ff6557554f82b919c38fb8a089b03c
QSI-2026-12045	tool_output	OSV.dev RUSTSEC-2024-0436 (manifest)	ea867adb8ba26c96c3779becf55b3de25d46f37f23ab55131e09c98a8b0
QSI-2026-12046	tool_output	OSV.dev RUSTSEC-2026-0173 (manifest)	0ecc182141bd9effc6d35bc24dc20c57e6473fb3877ad59a5fbdb1cd55b

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

- Classical only — threshold signing and MPC protocols: Pedersen commitment (4 file(s))

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
threshold signing and MPC protocols	Pedersen commitment (4)	QSI-2026-12031	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys and shares must stay valid for how long
transaction and message signing	ECDSA (15), BLS12-381 (4)	QSI-2026-12032, QSI-2026-12033	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count

weak classically	2
Grover: acceptable to NIST, below ECCG/ANSSI guidance	7
n/a	112
vulnerable (Shor)	26

Detailed findings: QSI-2026-12031, QSI-2026-12032, QSI-2026-12033

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — discrete logarithms in finite fields Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (4)	Breaks finite-field Diffie–Hellman and DSA at every size. Move key establishment to ML-KEM (hybrid with X25519 or ECDH during the transition) and signatures to ML-DSA or SLH-DSA.
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (4), ECDSA (18), Pedersen commitment (4)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	BLAKE2 (2), Keccak-256 (1), RIPEMD-160 (2), SHA-2 (4)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	BLAKE2 (2), Keccak-256 (1), RIPEMD-160 (2), SHA-2 (4)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-12058 · P2 · 30 days

root-cause fix

Upgrade rustls-webpki to a fixed version (0.103.13, 0.104.0-alpha.7).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104).

QSI-2026-12062 · P2 · 30 days

root-cause fix

Upgrade rustls-webpki to a fixed version (0.103.13, 0.104.0-alpha.7).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-82j2-j2ch-gfr8 (CVE-2026-93599, RUSTSEC-2026-0104).

QSI-2026-12067 · P2 · 30 days

root-cause fix

Upgrade serde_with to a fixed version (3.21.0).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-7gcf-g7xr-8hxj.

QSI-2026-12038 · P2 · 30 days

root-cause fix

Upgrade bytes to a fixed version (1.11.1).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.

Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-434x-w66g-qw3r (CVE-2026-25541, RUSTSEC-2026-0007).

QSI-2026-12049 · P2 · 30 days

root-cause fix

Upgrade rsa to a fixed version (none published yet; see the advisory for mitigations).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-4grx-2x9w-596c (CVE-2023-49092, GHSA-c38w-74pg-36hr, RUSTSEC-2023-0071).

QSI-2026-12051 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346).

QSI-2026-12054 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.45).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

	Dependency scan no longer reports GHSA-2mjx-qc3c-rqvc (RUSTSEC-2026-0285).

QSI-2026-12055 · P2 · 30 days root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-c8m5-c952-6cm4 (RUSTSEC-2026-0346).

QSI-2026-12064 · P2 · 30 days root-cause fix

Upgrade rustls-webpki to a fixed version (0.103.10, 0.104.0-alpha.5).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-pwjx-qhcg-rvj4 (CVE-2026-93602, RUSTSEC-2026-0049).

QSI-2026-12047 · P2 · 30 days root-cause fix

Upgrade rand to a fixed version (0.10.1, 0.8.6, 0.9.3).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097).

QSI-2026-12048 · P2 · 30 days root-cause fix

Upgrade rand to a fixed version (0.10.1, 0.8.6, 0.9.3).

Rationale	Eliminates the root cause described in the finding.

Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-cq8v-f236-94qc (RUSTSEC-2026-0097).

QSI-2026-12050 · P2 · 30 days

root-cause fix

Upgrade rsa to a fixed version (0.9.10).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-9c48-w39g-hm26 (CVE-2026-21895).

QSI-2026-12052 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-h2v8-394m-p7vv.

QSI-2026-12053 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

	Dependency scan no longer reports GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349).

QSI-2026-12057 · P2 · 30 days

root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-w44r-93j7-qv28 (RUSTSEC-2026-0349).

QSI-2026-12059 · P2 · 30 days

root-cause fix

Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098).

QSI-2026-12060 · P2 · 30 days

root-cause fix

Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099).

QSI-2026-12063 · P2 · 30 days

root-cause fix

Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).

Rationale	Eliminates the root cause described in the finding.

Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-965h-392x-2mh5 (CVE-2026-93600, RUSTSEC-2026-0098).

QSI-2026-12065 · P2 · 30 days root-cause fix

Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-xgp8-3hg3-c2mh (CVE-2026-93601, RUSTSEC-2026-0099).

QSI-2026-12043 · P2 · 30 days root-cause fix

Upgrade keccak to a fixed version (0.1.6).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-3288-p39f-rqpv (RUSTSEC-2026-0012).

QSI-2026-12056 · P2 · 30 days root-cause fix

Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.

Closure criteria	Dependency scan no longer reports GHSA-h2v8-394m-p7vv (RUSTSEC-2026-0348).

QSI-2026-12036 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0190, or the triage is recorded.

QSI-2026-12039 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0204; upgrade crossbeam-epoch to a fixed version (0.9.20) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0204, or the triage is recorded.

QSI-2026-12040 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0387, or the triage is recorded.

QSI-2026-12041 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0385; upgrade ctutils to a fixed version (0.4.4) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0385, or the triage is recorded.

QSI-2026-12042 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0221; upgrade event-listener to a fixed version (5.4.2) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0221, or the triage is recorded.

QSI-2026-12044 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0186; upgrade memmap2 to a fixed version (0.9.11) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0186, or the triage is recorded.

QSI-2026-12061 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0382 (CVE-2026-111943); upgrade rustls-webpki to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.

Rationale	Eliminates the root cause described in the finding.

Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0382 (CVE-2026-111943), or the triage is recorded.

QSI-2026-12066 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0382 (CVE-2026-111943); upgrade rustls-webpki to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0382 (CVE-2026-111943), or the triage is recorded.

QSI-2026-12069 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0381 (CVE-2026-111943); upgrade webpki-roots to a fixed version (0.26.11, 1.0.10) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0381 (CVE-2026-111943), or the triage is recorded.

QSI-2026-12070 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0381 (CVE-2026-111943); upgrade webpki-roots to a fixed version (0.26.11, 1.0.10) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.

Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0381 (CVE-2026-111943), or the triage is recorded.

QSI-2026-12071 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0395; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0395, or the triage is recorded.

QSI-2026-12072 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0397; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0397, or the triage is recorded.

QSI-2026-12073 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0399; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

	Dependency scan no longer reports RUSTSEC-2026-0399, or the triage is recorded.

QSI-2026-12074 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0400; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0400, or the triage is recorded.

QSI-2026-12075 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0401; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0401, or the triage is recorded.

QSI-2026-12076 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0402; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0402, or the triage is recorded.

QSI-2026-12077 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0403; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0403, or the triage is recorded.

QSI-2026-12078 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0404; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0404, or the triage is recorded.

QSI-2026-12079 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0406; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0406, or the triage is recorded.

QSI-2026-12080 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0410; upgrade zstd-sys to a fixed version (2.1.2) if it applies.

Rationale	Eliminates the root cause described in the finding.

Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0410, or the triage is recorded.

QSI-2026-12068 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0386; upgrade tokio to a fixed version (1.51.6, 1.53.3) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0386, or the triage is recorded.

QSI-2026-12031 · P3 · 90 days

short-term risk reduction

Add Pedersen commitment usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12032 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.

Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12033 · P3 · 90 days

short-term risk reduction

Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-12031 · P4 · 90 days

root-cause fix

Plan migration of Pedersen commitment to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-12032 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.

Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-12033 · P4 · 90 days root-cause fix

Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-12081 · P4 · 90 days root-cause fix

Update the development and build dependencies listed in the evidence.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Tooling upgrades may change build output; compare release artefacts.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	The dependency scan reports no advisories for tooling packages, or the remaining ones are recorded.

QSI-2026-12035 · P4 · 90 days root-cause fix

Replace adler with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	adler replaced, or the decision recorded.

QSI-2026-12037 · P4 · 90 days root-cause fix

Replace bincode with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	bincode replaced, or the decision recorded.

QSI-2026-12045 · P4 · 90 days root-cause fix

Replace paste with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	paste replaced, or the decision recorded.

QSI-2026-12046 · P4 · 90 days root-cause fix

Replace proc-macro-error2 with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	proc-macro-error2 replaced, or the decision recorded.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P2	QSI-2026-12058	Upgrade rustls-webpki to a fixed version (0.103.13, 0.104.0-alpha.7).	root-cause fix
P2	QSI-2026-12062	Upgrade rustls-webpki to a fixed version (0.103.13, 0.104.0-alpha.7).	root-cause fix
P2	QSI-2026-12067	Upgrade serde_with to a fixed version (3.21.0).	root-cause fix
P2	QSI-2026-12038	Upgrade bytes to a fixed version (1.11.1).	root-cause fix
P2	QSI-2026-12049	Upgrade rsa to a fixed version (none published yet; see the advisory for mitigations).	root-cause fix
P2	QSI-2026-12051	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix
P2	QSI-2026-12054	Upgrade rustls to a fixed version (0.23.45).	root-cause fix
P2	QSI-2026-12055	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix
P2	QSI-2026-12064	Upgrade rustls-webpki to a fixed version (0.103.10, 0.104.0-alpha.5).	root-cause fix
P2	QSI-2026-12047	Upgrade rand to a fixed version (0.10.1, 0.8.6, 0.9.3).	root-cause fix
P2	QSI-2026-12048	Upgrade rand to a fixed version (0.10.1, 0.8.6, 0.9.3).	root-cause fix
P2	QSI-2026-12050	Upgrade rsa to a fixed version (0.9.10).	root-cause fix
P2	QSI-2026-12052	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix
P2	QSI-2026-12053	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix
P2	QSI-2026-12057	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix
P2	QSI-2026-12059	Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).	root-cause fix
P2	QSI-2026-12060	Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).	root-cause fix
P2	QSI-2026-12063	Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).	root-cause fix
P2	QSI-2026-12065	Upgrade rustls-webpki to a fixed version (0.103.12, 0.104.0-alpha.6).	root-cause fix

P2	QSI-2026-12043	Upgrade keccak to a fixed version (0.1.6).	root-cause fix
P2	QSI-2026-12056	Upgrade rustls to a fixed version (0.23.46, 0.24.0-dev.2).	root-cause fix

60 days

Priority	ID	Action	Type
P3	QSI-2026-12036	Triage RUSTSEC-2026-0190; upgrade anyhow to a fixed version (1.0.103) if it applies.	root-cause fix
P3	QSI-2026-12039	Triage RUSTSEC-2026-0204; upgrade crossbeam-epoch to a fixed version (0.9.20) if it applies.	root-cause fix
P3	QSI-2026-12040	Triage RUSTSEC-2026-0387; upgrade crypto-bigint to a fixed version (0.7.6) if it applies.	root-cause fix
P3	QSI-2026-12041	Triage RUSTSEC-2026-0385; upgrade ctutils to a fixed version (0.4.4) if it applies.	root-cause fix
P3	QSI-2026-12042	Triage RUSTSEC-2026-0221; upgrade event-listener to a fixed version (5.4.2) if it applies.	root-cause fix
P3	QSI-2026-12044	Triage RUSTSEC-2026-0186; upgrade memmap2 to a fixed version (0.9.11) if it applies.	root-cause fix
P3	QSI-2026-12061	Triage RUSTSEC-2026-0382 (CVE-2026-111943); upgrade rustls-webpki to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.	root-cause fix
P3	QSI-2026-12066	Triage RUSTSEC-2026-0382 (CVE-2026-111943); upgrade rustls-webpki to a fixed version (0.103.16, 0.104.0-alpha.8) if it applies.	root-cause fix
P3	QSI-2026-12069	Triage RUSTSEC-2026-0381 (CVE-2026-111943); upgrade webpki-roots to a fixed version (0.26.11, 1.0.10) if it applies.	root-cause fix
P3	QSI-2026-12070	Triage RUSTSEC-2026-0381 (CVE-2026-111943); upgrade webpki-roots to a fixed version (0.26.11, 1.0.10) if it applies.	root-cause fix
P3	QSI-2026-12071	Triage RUSTSEC-2026-0395; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12072	Triage RUSTSEC-2026-0397; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12073	Triage RUSTSEC-2026-0399; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12074	Triage RUSTSEC-2026-0400; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12075	Triage RUSTSEC-2026-0401; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12076	Triage RUSTSEC-2026-0402; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12077	Triage RUSTSEC-2026-0403; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix

P3	QSI-2026-12078	Triage RUSTSEC-2026-0404; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12079	Triage RUSTSEC-2026-0406; upgrade zstd-safe to a fixed version (7.4.0, 8.2.0) if it applies.	root-cause fix
P3	QSI-2026-12080	Triage RUSTSEC-2026-0410; upgrade zstd-sys to a fixed version (2.1.2) if it applies.	root-cause fix
P3	QSI-2026-12068	Triage RUSTSEC-2026-0386; upgrade tokio to a fixed version (1.51.6, 1.53.3) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type
P3	QSI-2026-12031	Add Pedersen commitment usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12032	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-12033	Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-12031	Plan migration of Pedersen commitment to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12032	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12033	Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-12081	Update the development and build dependencies listed in the evidence.	root-cause fix
P4	QSI-2026-12035	Replace adler with a maintained alternative or record why it stays.	root-cause fix
P4	QSI-2026-12037	Replace bincode with a maintained alternative or record why it stays.	root-cause fix
P4	QSI-2026-12045	Replace paste with a maintained alternative or record why it stays.	root-cause fix
P4	QSI-2026-12046	Replace proc-macro-error2 with a maintained alternative or record why it stays.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-12058	Re-run the dependency scan after upgrading.
QSI-2026-12062	Re-run the dependency scan after upgrading.
QSI-2026-12067	Re-run the dependency scan after upgrading.
QSI-2026-12031	Re-run the inventory after refactoring.
QSI-2026-12032	Re-run the inventory after refactoring.
QSI-2026-12033	Re-run the inventory after refactoring.
QSI-2026-12038	Re-run the dependency scan after upgrading.
QSI-2026-12049	Re-run the dependency scan after upgrading.
QSI-2026-12051	Re-run the dependency scan after upgrading.
QSI-2026-12054	Re-run the dependency scan after upgrading.
QSI-2026-12055	Re-run the dependency scan after upgrading.
QSI-2026-12064	Re-run the dependency scan after upgrading.
QSI-2026-12047	Re-run the dependency scan after upgrading.
QSI-2026-12048	Re-run the dependency scan after upgrading.
QSI-2026-12050	Re-run the dependency scan after upgrading.
QSI-2026-12052	Re-run the dependency scan after upgrading.
QSI-2026-12053	Re-run the dependency scan after upgrading.
QSI-2026-12057	Re-run the dependency scan after upgrading.
QSI-2026-12059	Re-run the dependency scan after upgrading.
QSI-2026-12060	Re-run the dependency scan after upgrading.
QSI-2026-12063	Re-run the dependency scan after upgrading.
QSI-2026-12065	Re-run the dependency scan after upgrading.
QSI-2026-12043	Re-run the dependency scan after upgrading.
QSI-2026-12056	Re-run the dependency scan after upgrading.
QSI-2026-12036	Re-run the dependency scan after upgrading.
QSI-2026-12039	Re-run the dependency scan after upgrading.
QSI-2026-12040	Re-run the dependency scan after upgrading.
QSI-2026-12041	Re-run the dependency scan after upgrading.
QSI-2026-12042	Re-run the dependency scan after upgrading.
QSI-2026-12044	Re-run the dependency scan after upgrading.

QSI-2026-12061	Re-run the dependency scan after upgrading.
QSI-2026-12066	Re-run the dependency scan after upgrading.
QSI-2026-12069	Re-run the dependency scan after upgrading.
QSI-2026-12070	Re-run the dependency scan after upgrading.
QSI-2026-12071	Re-run the dependency scan after upgrading.
QSI-2026-12072	Re-run the dependency scan after upgrading.
QSI-2026-12073	Re-run the dependency scan after upgrading.
QSI-2026-12074	Re-run the dependency scan after upgrading.
QSI-2026-12075	Re-run the dependency scan after upgrading.
QSI-2026-12076	Re-run the dependency scan after upgrading.
QSI-2026-12077	Re-run the dependency scan after upgrading.
QSI-2026-12078	Re-run the dependency scan after upgrading.
QSI-2026-12079	Re-run the dependency scan after upgrading.
QSI-2026-12080	Re-run the dependency scan after upgrading.
QSI-2026-12068	Re-run the dependency scan after upgrading.
QSI-2026-12081	Re-run the dependency scan after updating.
QSI-2026-12034	Re-scan after upgrades.
QSI-2026-12035	Re-run the dependency scan after the change.
QSI-2026-12037	Re-run the dependency scan after the change.
QSI-2026-12045	Re-run the dependency scan after the change.
QSI-2026-12046	Re-run the dependency scan after the change.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.2.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.3.0
inventory_document	1.1.0
solidity	1.3.0
source_code	1.4.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.3.0
OSV.dev API	v1

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Reviewed: 2026-10-13

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27. OSV advisory GHSA-965h-392x-2mh5 — OSV.dev. https://osv.dev/vulnerability/GHSA-965h-392x-2mh5 (modified 2026-09-19T03:55:32.245575348Z)
28. OSV advisory GHSA-9c48-w39g-hm26 — OSV.dev. https://osv.dev/vulnerability/GHSA-9c48-w39g-hm26 (modified 2026-09-10T03:50:32.802096391Z)
29. OSV advisory GHSA-c8m5-c952-6cm4 — OSV.dev. https://osv.dev/vulnerability/GHSA-c8m5-c952-6cm4 (modified 2026-10-27T16:15:05.354395696Z)
30. OSV advisory GHSA-cq8v-f236-94qc — OSV.dev. https://osv.dev/vulnerability/GHSA-cq8v-f236-94qc (modified 2026-09-10T03:50:43.958966498Z)
31. OSV advisory GHSA-h2v8-394m-p7vv — OSV.dev. https://osv.dev/vulnerability/GHSA-h2v8-394m-p7vv (modified 2026-10-27T16:15:05.337819489Z)
32. OSV advisory GHSA-pwjx-qhcg-rvj4 — OSV.dev. https://osv.dev/vulnerability/GHSA-pwjx-qhcg-rvj4 (modified 2026-09-19T03:55:49.529651638Z)
33. OSV advisory GHSA-w44r-93j7-qv28 — OSV.dev. https://osv.dev/vulnerability/GHSA-w44r-93j7-qv28 (modified 2026-10-27T16:15:05.340347056Z)
34. OSV advisory GHSA-xgp8-3hg3-c2mh — OSV.dev. https://osv.dev/vulnerability/GHSA-xgp8-3hg3-c2mh (modified 2026-09-19T03:55:49.146112754Z)
35. OSV advisory RUSTSEC-2024-0436 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2024-0436 (modified 2026-01-27T22:03:15.694766Z)
36. OSV advisory RUSTSEC-2025-0056 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2025-0056 (modified 2025-09-05T09:05:48Z)
37. OSV advisory RUSTSEC-2025-0141 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2025-0141 (modified 2026-01-16T11:07:28.867553Z)
38. OSV advisory RUSTSEC-2026-0173 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0173 (modified 2026-06-07T21:00:04.318567692Z)
39. OSV advisory RUSTSEC-2026-0186 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0186 (modified 2026-06-22T18:15:03.600280397Z)
40. OSV advisory RUSTSEC-2026-0190 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0190 (modified 2026-06-29T14:15:03.650725453Z)
41. OSV advisory RUSTSEC-2026-0204 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0204 (modified 2026-07-06T22:30:04.159269138Z)
42. OSV advisory RUSTSEC-2026-0221 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0221 (modified 2026-07-31T12:15:04.629484145Z)
43. OSV advisory RUSTSEC-2026-0381 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0381 (modified 2026-10-29T10:55:54.096394276Z)
44. OSV advisory RUSTSEC-2026-0382 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0382 (modified 2026-10-29T10:55:54.096394276Z)
45. OSV advisory RUSTSEC-2026-0385 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0385 (modified 2026-10-23T08:30:02.389575143Z)
46. OSV advisory RUSTSEC-2026-0386 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0386 (modified 2026-10-23T14:45:03.847864944Z)
47. OSV advisory RUSTSEC-2026-0387 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0387 (modified 2026-10-24T03:45:02.365137418Z)
48. OSV advisory RUSTSEC-2026-0395 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0395 (modified 2026-11-02T09:30:03.420559993Z)
49. OSV advisory RUSTSEC-2026-0397 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0397 (modified 2026-11-02T09:30:03.445177143Z)

50. OSV advisory RUSTSEC-2026-0399 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0399 (modified 2026-11-02T09:30:03.328955732Z)
51. OSV advisory RUSTSEC-2026-0400 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0400 (modified 2026-11-02T09:30:03.428341570Z)
52. OSV advisory RUSTSEC-2026-0401 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0401 (modified 2026-11-02T09:30:03.353698025Z)
53. OSV advisory RUSTSEC-2026-0402 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0402 (modified 2026-11-02T09:30:04.162155944Z)
54. OSV advisory RUSTSEC-2026-0403 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0403 (modified 2026-11-02T09:30:04.136442771Z)
55. OSV advisory RUSTSEC-2026-0404 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0404 (modified 2026-11-02T09:30:04.168804805Z)
56. OSV advisory RUSTSEC-2026-0406 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0406 (modified 2026-11-02T09:30:03.591194710Z)
57. OSV advisory RUSTSEC-2026-0410 — OSV.dev. https://osv.dev/vulnerability/RUSTSEC-2026-0410 (modified 2026-10-26T14:00:05.788985783Z)
58. OSV.dev — open-source vulnerability database and API — Open Source Security Foundation. https://osv.dev/ (the database QSI queries for each exactly pinned version; every advisory id in a finding resolves at https://osv.dev/vulnerability/<id>)
59. P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997) — SIAM. https://doi.org/10.1137/S0097539795293172 (peer-reviewed article)
60. R. Babbush et al. — Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (arXiv:2603.28846) — arXiv. https://arxiv.org/abs/2603.28846 (preprint (2026-03-30, v2 2026-04-15): 256-bit ECDLP with < 1200 logical qubits and < 90 M Toffoli gates (or < 1450 and < 70 M); < 500,000 physical qubits at 10^-3 error, minutes)
61. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
RIPEMD-160	RIPEMD-160	weak classically	legacy	2	2	zcash_transparent/src/pczt/updater.rs:3; zcash_transparent/src/util/hash160.rs:4
BLAKE2		Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	2	2	components/equihash/src/blake2b.rs; components/equihash/tromp/blake2b.h
Keccak-256	Keccak-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	1	1	components/eip681/src/parse.rs:15
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	4	4	zcash_primitives/src/block.rs:124; zcash_transparent/src/pczt/updater.rs:98; zcash_transparent/src/util/hash160.rs:29
Crypto library jubjub (jubjub)	0.11.1	n/a	acceptable	1	1	Cargo.lock
Crypto library p384 (RustCrypto elliptic curves)	0.13.0	n/a	acceptable	1	1	Cargo.lock
Crypto library p256 (RustCrypto elliptic curves)	0.13.2	n/a	acceptable	1	1	Cargo.lock
Crypto library bellman (bellman)	0.15.0	n/a	acceptable	1	1	Cargo.lock
Crypto library orchard (Zcash protocol crates)	0.16.0	n/a	acceptable	1	1	Cargo.lock
Crypto library ecdsa (python-ecdsa)	0.16.9	n/a	acceptable	1	1	Cargo.lock

Crypto library ring (ring (Rust))	0.17.14	n/a	acceptable	1	1	Cargo.lock
Crypto library halo2_poseidon (halo2 (Zcash))	0.2.0	n/a	acceptable	1	1	Cargo.lock
Crypto library sinsemilla (sinsemilla)	0.2.0	n/a	acceptable	1	1	Cargo.lock
Crypto library zcash_spec (Zcash protocol crates)	0.2.1	n/a	acceptable	1	1	Cargo.lock
Crypto library rustls (rustls)	0.21.12	n/a	acceptable	1	1	Cargo.lock
Crypto library rustls (rustls)	0.23.25	n/a	acceptable	1	1	Cargo.lock
Crypto library zip32 (Zcash protocol crates)	0.3.0	n/a	acceptable	1	1	Cargo.lock
Crypto library secp256k1 (secp256k1 bindings)	0.33.1	n/a	acceptable	1	1	Cargo.lock
Crypto library halo2_proofs (halo2 (Zcash))	0.4.0	n/a	acceptable	1	1	Cargo.lock
Crypto library zcash_note_encryption (Zcash protocol crates)	0.5.1	n/a	acceptable	2	2	Cargo.lock; Cargo.toml
Crypto library halo2_gadgets (halo2 (Zcash))	0.6.0	n/a	acceptable	1	1	Cargo.lock
Crypto library pasta_curves (pasta_curves (Pallas/Vesta))	0.6.0	n/a	acceptable	1	1	Cargo.lock
Crypto library reddsa (reddsa)	0.6.1	n/a	acceptable	1	1	Cargo.lock
Crypto library bls12_381 (bls12_381)	0.9.0	n/a	acceptable	1	1	Cargo.lock
Crypto library sapling-crypto (Zcash protocol crates)	0.9.0	n/a	acceptable	1	1	Cargo.lock
Crypto library redjubjub (reddsa)	0.9.0	n/a	acceptable	1	1	Cargo.lock
Crypto library rsa (RustCrypto rsa)	0.9.6	n/a	acceptable	1	1	Cargo.lock

Crypto library x25519-dalek (ed25519-dalek)	2.0.1	n/a	acceptable	1	1	Cargo.lock
Crypto library ed25519-dalek (ed25519-dalek)	2.2.0	n/a	acceptable	1	1	Cargo.lock
Crypto library curve25519-dalek (ed25519-dalek)	4.1.3	n/a	acceptable	1	1	Cargo.lock
Crypto library jubjub (jubjub)	0.11	n/a	acceptable	1	1	Cargo.toml
Crypto library bellman (bellman)	0.15	n/a	acceptable	1	1	Cargo.toml
Crypto library orchard (Zcash protocol crates)	0.16	n/a	acceptable	1	1	Cargo.toml
Crypto library zcash_spec (Zcash protocol crates)	0.2	n/a	acceptable	1	1	Cargo.toml
Crypto library zip32 (Zcash protocol crates)	0.3	n/a	acceptable	1	1	Cargo.toml
Crypto library secp256k1 (secp256k1 bindings)	0.33	n/a	acceptable	1	1	Cargo.toml
Crypto library pasta_curves (pasta_curves (Pallas/ Vesta))	0.6	n/a	acceptable	1	1	Cargo.toml
Crypto library bls12_381 (bls12_381)	0.9	n/a	acceptable	1	1	Cargo.toml
Crypto library redjubjub (reddsa)	0.9	n/a	acceptable	1	1	Cargo.toml
Crypto library sapling-crypto (Zcash protocol crates)	0.9	n/a	acceptable	1	1	Cargo.toml
Crypto library zcash_protocol (Zcash protocol crates)	=0.11.0-pre.1	n/a	acceptable	1	1	Cargo.toml
Crypto library zcash_keys (Zcash protocol crates)	=0.17.0-pre.2	n/a	acceptable	1	1	Cargo.toml
Crypto library zcash_client_backend (Zcash protocol crates)	=0.25.0-pre.2	n/a	acceptable	1	1	Cargo.toml

Crypto library zcash_primitives (Zcash protocol crates)	=0.31.0-pre.2	n/a	acceptable	1	1	Cargo.toml
Crypto library zcash_proofs (Zcash protocol crates)	=0.31.0-pre.2	n/a	acceptable	1	1	Cargo.toml
Crypto library zcash_protocol (Zcash protocol crates)	workspace	n/a	acceptable	10	10	components/zcash_address/Cargo.toml; components/zip321/Cargo.toml; pczt/Cargo.toml
Crypto library bls12_381 (bls12_381)	workspace	n/a	acceptable	6	6	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library pasta_curves (pasta_curves (Pallas/Vesta))	workspace	n/a	acceptable	3	3	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library secp256k1 (secp256k1 bindings)	workspace	n/a	acceptable	6	6	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library zcash_proofs (Zcash protocol crates)	workspace	n/a	acceptable	3	3	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library zip32 (Zcash protocol crates)	workspace	n/a	acceptable	8	8	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library zcash_note_encryption (Zcash protocol crates)	workspace	n/a	acceptable	4	4	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library zcash_primitives (Zcash protocol crates)	workspace	n/a	acceptable	6	6	pczt/Cargo.toml; zcash/Cargo.toml; zcash_client_backend/Cargo.toml
Crypto library redjubjub (reddsa)	workspace	n/a	acceptable	4	4	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_primitives/Cargo.toml

Crypto library jubjub (jubjub)	workspace	n/a	acceptable	6	6	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library orchard (Zcash protocol crates)	workspace	n/a	acceptable	7	7	pczt/Cargo.toml; zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml
Crypto library zcash_keys (Zcash protocol crates)	workspace	n/a	acceptable	3	3	zcash_client_backend/Cargo.toml; zcash_client_sqlite/Cargo.toml; zcash_pool_migration/Cargo.toml
Crypto library zcash_client_backend (Zcash protocol crates)	workspace	n/a	acceptable	2	2	zcash_client_sqlite/Cargo.toml; zcash_pool_migration/Cargo.toml
Crypto library bellman (bellman)	workspace	n/a	acceptable	1	1	zcash_proofs/Cargo.toml
Crypto library zcash_spec (Zcash protocol crates)	workspace	n/a	acceptable	1	1	zcash_transparent/Cargo.toml
BLS12-381		vulnerable (Shor)	acceptable	4	4	zcash_proofs/src/circuit/sprout/mod.rs:340; zcash_proofs/src/lib.rs:15; zcash_proofs/src/prover.rs:3
ECDSA	secp256k1	vulnerable (Shor)	acceptable	15	15	pczt/src/roles/signer/mod.rs:278; zcash_client_backend/src/data_api.rs:3935; zcash_client_backend/src/data_api/wallet.rs:1380
ECDSA		vulnerable (Shor)	acceptable	3	3	pczt/src/roles/signer/mod.rs:299; zcash_transparent/src/builder.rs:153; zcash_transparent/src/pczt/signer.rs:171
Pedersen commitment		vulnerable (Shor)	acceptable	4	4	pczt/src/orchard.rs:23; zcash_primitives/src/transaction/components/orchard.rs:17; zcash_primitives/src/transaction/components/sapling.rs:14

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 03b3c2491116445855ca64e958f067d951c66e342cfeed96f84e97c91650ed99

C. Input files

File	Size	SHA-256
.cargo/config.toml	27	c50d9a3b08fa6bb6b71d5d4f7af76b8a61fbd04efd207e60ee78b4400789
.github/helpers/check-dep-graph.py	3784	b286aec9a590ce0319f4aee11cc48cbce887f92febf41741cdf90d0fa299
.github/helpers/check-release-tags.py	6021	1b89b7b1cb9e41b0e8d4b3d5979ca69ef31270418b39ee03aea1bc8aaa79
Cargo.lock	187600	f218ea41f705a0d7db4156816fd1c6ecd95828064cc198cd489ae1b38822
Cargo.toml	8344	2e9608a94f0314790472b62a27eaee56af1c42aa85eb1d386be211eee6bd
components/eip681/Cargo.toml	999	1b6ffda755504c497286d2e03243e7e3daa49e71ac6266c6e4949198d5b2
components/eip681/src/error.rs	9320	2737ae8953a208f2dfb3dfaa0f76748e3a43f116f6b974191e38e86f80a6
components/eip681/src/lib.rs	2572	7980128f201cc345395dcb045518c797bf19941f57dd6975ca9212027df4
components/eip681/src/parse.rs	76946	fee3ec17332402f6c9fe96c9b68b3008215a027a8efd85f7d13b568712c2
components/eip681/src/request.rs	29780	12837d313debdad012e56a171c9bd8c75f5a667fe54ab4d987ef47b6b0f6
components/equihash/Cargo.toml	952	34a0876441a88a58f9fb7fb11b1b83aa10c2889f0c8c5abc0a0adeb16f5e
components/equihash/build.rs	433	2716a2cf3d85f116d8873b231b3fc19c3f9cc8aad95063592041767aab1a
components/equihash/src/blake2b.rs	1858	f7bf12ab1ddfc2693c6e7594cf00f77581a278b178ba9d8a144715faa384
components/equihash/src/lib.rs	1619	4ec356326258ec10ed628cb4f481c2a8efbe79c374b8ec094ac0b6f98fcf
components/equihash/src/minimal.rs	9536	74bec30cd45cb772422bd76e4144774a880792e7ef6d59bc4e12b86e40d8
components/equihash/src/params.rs	1302	c77e235d106c1ad0b5a130ba3f9c54bed8b99d061f7dcaab4b1e9424ec03
components/equihash/src/tromp.rs	9202	f5843bdfd4f3a6eac065554f69476346790906856 4f259b1ccb226fd3c1c
components/equihash/src/verify.rs	9595	1d2398f57391481d9f10030aadcbbc7a3230e0b0cbaabfcfd9793faa71ce
components/equihash/tromp/blake2b.h	1676	1006c3a6361483718cc7ba404b9078f233ea59a04887330902076c3ed1bf
components/equihash/tromp/equi.h	1100	94f0820ac42fe5ecfeaee857649afecf43d465068461bd4efb699050ed1c
components/equihash/tromp/equi_miner.c	22553	212d91f704d56a7faf071d41ffe88b0c98ea10ef0ccd1fe0f806fd88bc5f
components/equihash/tromp/portable_endian.h	3176	e60f79c27af38844051bd9a62dcf2bca6a9131ee005959bf5919bb1fca2f
components/f4jumble/Cargo.toml	876	54ced6c2c71f577c9d6bebca57ee26dece422b937e2c0e659d38b37c31cf
components/f4jumble/src/lib.rs	13874	c860d1b3b199cf5b5dc57eb63eca5c500a0d11a917d80a618477e0fba90d
components/payment_uri/Cargo.toml	764	dafd5537e16bbad92e64108de3479cef376374319e215796a30cc5bd1889
components/payment_uri/src/bitcoin.rs	8825	27c03e6863c0b591ed0f675e8c251ab7d1503dc2d2190e6389bb1ec950d7

components/payment_uri/src/lib.rs	15178	4d9b7fff3027fe4d40904b6df7cde56aa492920258658c7022324b618ba4
components/payment_uri/src/solana.rs	7304	11c3159e45160c5f8090f988adac744e8827e304898f696eb1b769fb692b
components/zcash_address/Cargo.toml	1089	9db2888c15565750230a1798ecdd40666d511f27e82a40927e8161ee1853
components/zcash_address/src/constants.rs	15691	542102eb1f87bc3746fdfc1f5d85fc3f26dafcec4ef32ec00e76655a9c3e
components/zcash_address/src/convert.rs	12197	460a92b779817ceb1e51b8b4acbfcdc9e1d807f60a9284c5c042f8620827
components/zcash_address/src/encoding.rs	14821	d2df7640e37fd24677ff5ab31245507380f1ce876a71f00f927b6afe03c6
components/zcash_address/src/kind.rs	17	1e1cac26248df3cbe9959d287d59160893b5b7ae273e76d9e344d2947b0d
components/zcash_address/src/kind/unified.rs	48512	edfa017f1b7534670be00d0f76bfc44666399678f000b62985c359384334
components/zcash_address/src/kind/unified/address.rs	35355	8e22fd18ac1afd1a2688b2a6cb0bb36f74385d3226dff5d92d046b7ca8e0
components/zcash_address/src/kind/unified/fvk.rs	28030	cc3974080fe53b2974f98354902d11804d2f621395a1a0154546e47f4118
components/zcash_address/src/kind/unified/ivk.rs	23102	29d5c6ddfb5f57caa3e4659b02d0c1776c1f02851f9b291b970b9cfb9420
components/zcash_address/src/kind/unified/revision.rs	2363	15376624d2ee7decca87551d9f7e467b1e36ed5b15afa9311efea146fe9a
components/zcash_address/src/lib.rs	16793	a8ab232c059b661a295b258a132d9788952cf1806d83f75ff3bbef96031b
components/zcash_encoding/Cargo.toml	896	59fad1e11bce86da49aa97cc8b3ea7930f9cfe666d277deb7ae61051cd0a
components/zcash_encoding/src/lib.rs	19244	79cd47d3d73370fff5745c3e2efaeebc0104c0ec96aaf20cb085670829459
components/zcash_protocol/Cargo.toml	1605	fd684937f5ed041aea5c6b4e5f0562860a28b0a3598dcc713006adaba17d
components/zcash_protocol/src/consensus.rs	48506	004f0da79523cbe407a477cb7cc3abf0735f72946defd61176ef1dde6fd7
components/zcash_protocol/src/constants.rs	2645	92bf49080134e943f100280f4046684734be9b652f1fc4f1a7fa4c7a1862
components/zcash_protocol/src/constants/mainnet.rs	4714	bc5ec2e50adf3945f3d2a846d4b9922cd53de08ec5a9d7a1d2764624dbe0
components/zcash_protocol/src/constants/regtest.rs	5099	db11b07c4b20e28901fe7018ad513d24a645c271c4d441e8d9c21b373ad3
components/zcash_protocol/src/constants/testnet.rs	4774	a4a154d59cab94159754c42305419af94edbffc10b2fa2212b21806b675e
components/zcash_protocol/src/lib.rs	2867	9def3b5138d0eae8153fc730cb4f033b75b8a8db00313062d7cf470b03d5
components/zcash_protocol/src/local_consensus.rs	10334	05bb22840ac92f141fb5c8cfba7a77c0c53116a6f767ebb77f9a35847509
components/zcash_protocol/src/memo.rs	18347	24a4f6444e20add60ec0b558a52125b38f161c394e64b6b07f44757426af
components/zcash_protocol/src/txid.rs	4047	7d19917afcf7b8f98fee478e50411673aed1d03883a0691800420bf783ff
components/zcash_protocol/src/value.rs	19311	f44c8a8740df244cbd3f01ace6d8343382fa414d22a2b7269feb4fe264b6
components/zcash_protocol/src/zip318.rs	69613	635e281901e1814b8b17ceaa1ef0e20650b2824549205157537f4e7040a2
components/zip321/Cargo.toml	1121	94b27b0f4640c9c671af2254381665e755418946238218bec039b8699c40

components/zip321/src/lib.rs	51154	ed6f284170ad237f71b363cfeb5e5c45df7b997e6f68b3708326877219a5
pczt/Cargo.toml	4601	b2b58ab523aad8654013aa0bfb128511ec3cbba1383b3b5cd028901a889f
pczt/src/common.rs	22186	a58ee6e362805bf1f5534b5768639d9073b3f7798e73e49e9aa1554b1842
pczt/src/lib.rs	39598	e5547e561f4f22fd0bdb931b90f4728006b24a4eee2b45114a47290250f6
pczt/src/orchard.rs	103702	483324d3e983919c48ae0b7a94cbc7aec49d908fb328f4ea7a20210b4e66
pczt/src/roles.rs	20952	9cbf3067639b2a801f0fd8a3bba4a9c708a4041a337032923fa23b28fc3f
pczt/src/roles/combiner/mod.rs	3166	c44b30771ae5f2dd3ecf3ce1785fd50f9def83e6cb440d7c68ed89d708fa
pczt/src/roles/creator/mod.rs	16763	e5593763da9d0668322d4e68eabeb94012ea4a130bbf49f9576b8bebff8c
pczt/src/roles/io_finalizer/mod.rs	4820	8b706b8b04db961ba551c330127f0991bff99cb6a5a1d14c00c486aa86b5
pczt/src/roles/low_level_signer/mod.rs	12419	ece5222d7f186669afb88bef74abc52c9d4b5b0fea10396199cc07603923
pczt/src/roles/prover/mod.rs	1718	362b884b4d44938e07d904918259fa427f5262df751e53cf456dc4ab9383
pczt/src/roles/prover/orchard.rs	6566	10575343ec692fdc7af0e8127f750d0c497064ff7ba769d9bedd3d763e4d
pczt/src/roles/prover/sapling.rs	4252	188d4bad03cf4d0b310b61348c4288ac4c0658178d7a4854cd10015ceeae
pczt/src/roles/redactor/mod.rs	1437	e1e4d6d092720093205b2c9eeeeeac7b3f2443af4870c6826e88611b05e8
pczt/src/roles/redactor/orchard.rs	9162	3dadb7993fda81ecf4f2cf8c5ca33b04ed040c77a14a7b7a92f295b0fb2b
pczt/src/roles/redactor/sapling.rs	7324	1b359569364bb9079d22420e737cc96298943bde809a85ba10fa8a589e8f
pczt/src/roles/redactor/transparent.rs	6974	db5949deb4dd851de271f66246e51338385530fc644d5d7700bf30196978
pczt/src/roles/signer/batch.rs	17769	0227a2c6cc10c71bf6505869514fcd35f45de6a4514cc33732dbd488ad61
pczt/src/roles/signer/mod.rs	31821	2145651f5e27de716cc6e0a308e0ea879a7e900f912506ab7ebf546cdcd9
pczt/src/roles/spend_finalizer/mod.rs	2342	4e1cd75e16f8c27dc33fd79545d5104e902c73962ae94f5feeafde879955
pczt/src/roles/tx_extractor/mod.rs	5991	b7d097fe9de522d037d81c3b65af7efbd1cc3978a3fcd3a0fcbfe9d8e3c9
pczt/src/roles/tx_extractor/orchard.rs	1331	01f6047241666d2069c239f7bc76e5230c34ccba0ba0755d439954c75f8a
pczt/src/roles/tx_extractor/sapling.rs	870	54a870de9e89638820ba459bfec0e923e548280d29cc1f2df72a2ea501df
pczt/src/roles/tx_extractor/transparent.rs	534	6ef0d010a927cc969b6a9f78fff37eb046fc4bdd959e816e41c32033416f
pczt/src/roles/updater/mod.rs	14631	4a798f0c0e1b2fe48e127ec14fbf2dd30ae5f2fc45a7159e2ac43fa9649e
pczt/src/roles/updater/orchard.rs	2528	25d3c19fb3025e6be3a58bd785235f95071f0e3f110f73b3155896c4fc7e
pczt/src/roles/updater/sapling.rs	1235	9551f53c850e5e3909228144b1152b76b6d9858128d10a210cf4f4e51869
pczt/src/roles/updater/transparent.rs	1105	cc24f415a7f1a5ea143972a0b564ff3fd6f80697a6f7293deeacbdda0959
pczt/src/roles/verifier/mod.rs	846	04d4616e82c37edc7672d7b4346c6bc5cbf1e2d00fccaf301d717c8a4fb4
pczt/src/roles/verifier/orchard.rs	2527	f2c1e65e15176decca66927cbcaa3a38c3a09df08f78c4dc31f6f2d8682c
pczt/src/roles/verifier/sapling.rs	1310	818fb4ca3d242b4e35b55d8d26fb97b49879d12915e1cf8a432c70bb586f
pczt/src/roles/verifier/transparent.rs	1251	343696b07768a50068d924730e8869289f5489e04548e325060a0536de6e
pczt/src/sapling.rs	33117	404edf4e4b86fd4273e4dd092e899698c035dc381929388965246de9e7fe

pczt/src/transparent.rs	18048	c78e8f88ae58b2bb8fa91dcdb2ca35dbe3b964d6413e557e1a39a7963717
rust-toolchain.toml	66	33770f77c5cb48a231ed517388b85dda27873fceceaa52b18a06108af4fd
zcash/Cargo.toml	999	1b4b9b26c7729473798d483200452c80a4034661724804795bec208f8db1
zcash/src/lib.rs	360	8851e622aaaf23dabe44d82ad3224ace6016a91b4d07691d0794fbb9f249
zcash_client_backend/Cargo.toml	10453	4e22a4980fe4dc63bc1f339b3bc8d15a8ae4762f6f9d4d302170aec0d331
zcash_client_backend/build.rs	4290	59c2c56d8e84904759c0cf65ca603c63024339972f65fa43fa1e88612f7e
zcash_client_backend/lightwallet-protocol/walletrpc/compact_formats.proto	6490	d8d0c8aaa5ceec7d5dcc188ef25b04011df2ec0254901620fce982fc13ae
zcash_client_backend/lightwallet-protocol/walletrpc/service.proto	15232	80575dbe59a9bf2e6b79e2391eb78679c453f1a0477292eb97ea3b58bb6c
zcash_client_backend/proto/proposal.proto	7211	f72aeefdee46bc9b42504faf0b3f811b736289ca9f1812ee0230428e7e37
zcash_client_backend/src/data_api.rs	245109	c0000e72322a214d7eba1d298a1b1c349f0c37b305de02515389be22d35d
zcash_client_backend/src/data_api/anchor_retention.rs	18296	f9fe8ec75a850be553a2188601665a956d2f0ad7646892c65a1efbac0a1d
zcash_client_backend/src/data_api/chain.rs	29089	9626e0dd1824ed8d7a10d94fdedab843fa456703df8ed22908c1167c58dc
zcash_client_backend/src/data_api/chain/error.rs	1953	cfae1c314f0abf4a41b9c78e3e0055e87777bebdd1616be1e26de2169120
zcash_client_backend/src/data_api/defaults.rs	5487	a3d26ae57343909dbcc21dfe352200c7a0b1e4755cf78358f81982327c9f
zcash_client_backend/src/data_api/error.rs	24427	2056fc5d453204b6819866625a69e729e5ddb723545fec8660240ef1cfcf
zcash_client_backend/src/data_api/ll.rs	42402	62df4bc82b5bcc7e0f48c0c6541ee1e917a97d201072b805ed7ad6cc5fe1
zcash_client_backend/src/data_api/ll/wallet.rs	87759	4dae07184069adcc411e3a05761c2989f1280140466c6dde7fe21e202ed4
zcash_client_backend/src/data_api/locking.rs	25153	11ee7c5309193cada1774298165df61b984265735844066f923c66d205ba
zcash_client_backend/src/data_api/scanning.rs	6652	e65da16c7f09a9d70e93cba3c9a7773edd3d85a190b8e84d8033a2d4340b
zcash_client_backend/src/data_api/scanning/spanning_tree.rs	27619	01e603b5a42b5fe7387782571bb5e6ede18b078f6479b4db83f41a4c97a1
zcash_client_backend/src/data_api/wallet.rs	183539	ae75d6dfecc19bc4941254734822cab1b9bbf982c175257f61c6dea7f578
zcash_client_backend/src/data_api/wallet/input_selection.rs	139583	fd057ab405a33ac03b81a6c7e69dc91739e8263b75cfb415831d2238257
zcash_client_backend/src/data_api/zip318.rs	6604	f9c191c87e6c196b7d09c7f318e583e58604e421282de05826f9e4d860e2
zcash_client_backend/src/decrypt.rs	32717	b160542f3aae544086c06aa66f59a5cef4ec2edf3743dde57effa6f8a419
zcash_client_backend/src/fees.rs	33472	8744c008a8eded3dc0d0698ca970f7438f21ef0494faaf07a7e604ca29c3
zcash_client_backend/src/fees/common.rs	59203	45b6e67e13bc3ba691169b60a156cdbce2fa66091ac1726e991075441a8e
zcash_client_backend/src/fees/fixed.rs	10048	c937b76e00627ab3a0f11a99d808c896d408278a9cefc9b57fa54c90549f
zcash_client_backend/src/fees/orchard.rs	3398	a25ca7f8f8abef02ac9e65a07e3537dfebca881a8b44876180a5621f20fd
zcash_client_backend/src/fees/sapling.rs	2984	a32db08a0e1e0e8ded64f1565122b0b35325c3e6ebc2452119dcc3cfe462

zcash_client_backend/src/fees/standard.rs	949	941ad20e3bb39396b7c215a817ff8c12b9c195627b8d131d61a69810528f
zcash_client_backend/src/fees/zip317.rs	85127	6a371e5dbe1e623f38f823962b575d04b80b2070c959dc1df7279ec4ac3c
zcash_client_backend/src/lib.rs	5709	13a038a66f74ce48ae05234e5e2a257fcbef8e7819c3749380d89b739b0b
zcash_client_backend/src/proposal.rs	118472	e29c30b0ad345264d476cccd0922c90d40197de0c2184b4bbcc19e8cba86
zcash_client_backend/src/proto.rs	51915	e2b4494c29458019254d1ae59691db454e99ee4f67bf9407d82abc8d0666
zcash_client_backend/src/proto/compact_formats.rs	8097	d6672eb5c49b158bbe0614f161f4be292d3203fc3fe56dcda503e5c2769f
zcash_client_backend/src/proto/proposal.rs	11947	ec6933e5bf85f4492aaba37b0c7fddd6ae9d61c211fdd5d08bd21cfde5f9
zcash_client_backend/src/proto/service.rs	45200	f5058c7c262ba409fba58d27afd78e1ad7fe371b3834c72e2a18ef39d291
zcash_client_backend/src/scan.rs	21712	c68b7bec12c77989eb688d7b04b1a861406d5c43aba5121b260d36e89786
zcash_client_backend/src/scanning.rs	49247	ec0b5539e04cf07377f3b850c6fa47aa70aaf8b3d7b3af40e39888127cac
zcash_client_backend/src/scanning/compact.rs	52785	3a4d174fb58bf31bd66f18b1c1cd4f848469ddef7fc462315546cadc765d
zcash_client_backend/src/scanning/full.rs	46459	137dc8301d07063ccb5c652e6e603498e10cb64eb64d4ff0d5f94740a0d1
zcash_client_backend/src/serialization.rs	19	5aa43d739cce707b80c859f04cbe56d453d2eef47fbb15701ab261d980b4
zcash_client_backend/src/serialization/shardtree.rs	4157	5a428a638ac0e4675380527143aacee09882b1302d51921675cdcdcea154
zcash_client_backend/src/sync.rs	25733	5acf93c423c9ceab91d72e4569b41c77daa01a2faa17f1d102899661eadc
zcash_client_backend/src/sync/decryptor.rs	19471	e3b92c0069d6ff8af8926746e2e61308ce03c65e5eba51a650ba826744f8
zcash_client_backend/src/task.rs	737	8018642f411bba591196f3235d0eaa8bb2b38f3fa6624a144bbc07ab3923
zcash_client_backend/src/tor.rs	10717	9635522697a2597655e9edbf3642de711edb6b74939904a7a3b48cf7ce10
zcash_client_backend/src/tor/grpc.rs	5346	ebe82a7c198de14a7aa1b5a2a0378ef9b27abb8997b30134808f2344193b
zcash_client_backend/src/tor/http.rs	33664	95865c533750a44b372ac1f4c9c4c238524ed37b0ec0b22c0469412ba0df
zcash_client_backend/src/tor/http/cryptex.rs	8862	949045023c6219c056af17492297ca16fdf2e4477cf456eae9db54457d09
zcash_client_backend/src/tor/http/cryptex/binance.rs	1677	bd7a0ab892ac86cb794c02ca5555259cd8d7bc92281f597b5b4411b45ff2
zcash_client_backend/src/tor/http/cryptex/coin_ex.rs	1990	2c8aa1ea109d607595a2abdfb04098db1c93362f413a924591ccaa6a67ea
zcash_client_backend/src/tor/http/cryptex/coinbase.rs	1390	986ebfdb3397eb380b9eeba9378ddf3e20a33eb4270bd9d36902d7403d4a
zcash_client_backend/src/tor/http/cryptex/digi_finex.rs	1475	0f504c2ccc68f0613b96d7b84e26bc3fe0f47c2299a5b2200d15d115750c
zcash_client_backend/src/tor/http/cryptex/gemini.rs	1206	d2877b178c91daca5786de3afe57c5885dac40641791dce77b238a22475e
zcash_client_backend/src/tor/http/cryptex/kraken.rs	1764	2883622a7d89c98d16053f836505b974c1a541b8e041984964de849e8931

zcash_client_backend/src/tor/http/cryptex/ku_coin.rs	1652	50b04fa066a23dcfaf32b6c95077adf9e8666ed5413bd4f02a018779e3e4
zcash_client_backend/src/tor/http/cryptex/mexc.rs	1544	b523ae37dd21910cb8af633f5cbbb6df6e7c2096b65bbbbfb226c4a023a4
zcash_client_backend/src/tor/http/cryptex/xt.rs	1349	baab400a2acda0cfa68fe36671181a6bc40bdd3e6d02f5e5b685b1438585
zcash_client_backend/src/util.rs	2217	33b077d3b3b477696da054899610e02e20a9a30f5ce775d14deefdcbc8e8
zcash_client_backend/src/wallet.rs	52615	41e038af8afcdd9d7028619aa97c009c6be2ee08123a2d530b19f477b273
zcash_client_sqlite/Cargo.toml	8738	e004fc75fa8f0ca57c2b8697697409f9021faf01694992173b8bcd71ae59
zcash_client_sqlite/src/chain.rs	17888	448782a4c5939dfd1b09e820c38ddd5bd72a918117795c849cecd5f18fdb
zcash_client_sqlite/src/chain/init.rs	2034	e5494c0db48acfc86b71f8c2ad1519e0fbae9efc867fe999f68c368cbbb2
zcash_client_sqlite/src/chain/migrations.rs	74	130503e1c24a905ecc03e0adbebf703e751eb2b3a7120ade154e6d85cd5d
zcash_client_sqlite/src/chain/migrations/blockmeta.rs	1811	b8225c4718f9b8aa4cc49c29f70b1f5d8134a57570e22cd34c7e7f4971bb
zcash_client_sqlite/src/error.rs	28283	13be30df8619098d188304fc908b5b01a30fc33f0c6edfcd822041ed1460
zcash_client_sqlite/src/lib.rs	215065	c2517b23d072ff310ba22625dd394dc368f8549a1d9a1a279cf4553e2219
zcash_client_sqlite/src/pool_migration.rs	11321	64b16b055bb1c906722adea4c76ae069c4d19d92f156d6fd1880075266ab
zcash_client_sqlite/src/pool_migration/error.rs	6722	83a237affd5a30d98ff5a83af8b508151804fbee26876fa6db9334dd447b
zcash_client_sqlite/src/pool_migration/orchard_ironwood.rs	165551	f73e38bee4cf695144e7666e24c968505a1311eb9647a2a10c54412d7a17
zcash_client_sqlite/src/pool_migration/store.rs	100536	2db199ae15a7a84926abbb8e8e317cc00ce2ee76ba339a70c988649b8255
zcash_client_sqlite/src/util.rs	387	19e90bdf9de56be56cf2a6a00479e9856f78b674ad30e79954546874108f
zcash_client_sqlite/src/wallet.rs	349453	759b0d6c6a568b2a01a48049dddc38a06f1611e1dba52aded65232bcc14b
zcash_client_sqlite/src/wallet/commitment_tree.rs	79912	59ae465458b052defaf9684e1934c462211fb49a1f2f161cda59f3ffccac
zcash_client_sqlite/src/wallet/common.rs	53887	862e4652305a488cf95a41ba7912ef8729b90c0ff8b8bdffce307bf9c19f
zcash_client_sqlite/src/wallet/db.rs	102081	2adf12f943266a11f64bc908b3970c2d3bd6c8ead2e6947d763b0f2dad5a
zcash_client_sqlite/src/wallet/encoding.rs	13943	8e4d77d8bcc541be8653c238873ec8e5c64ed0c69cdda89cbd8738b9e5d7
zcash_client_sqlite/src/wallet/init.rs	67895	3c7f94680655eebc5d3a479826ba349f230986741ee5a3d1e1672a18a288
zcash_client_sqlite/src/wallet/init/migrations.rs	55309	9c15a8cb6bd81699461a87c39365f4634d8f10437945579c9b0b45a86c60
zcash_client_sqlite/src/wallet/init/migrations/account_delete_cascade.rs	31296	fa1c1ca1a2cbba3d5a3e0b03c0e0e829d27569816d479bb6c962dd114b72
zcash_client_sqlite/src/wallet/init/migrations/add_account_birthdays.rs	4347	310e3f68b83705ed58d9ec9a00f0085e8a3a382555b0874acc857ed80d0c
zcash_client_sqlite/src/wallet/init/migrations/add_account_uuids.rs	16228	131531f5d60981c620570d298093a7eae5cbfdc1cef6822a71c72c4b3993

zcash_client_sqlite/src/wallet/init/migrations/add_transaction_trust_marker.rs	8620	cd4f3b6e8c64a011fff7641ca218933b04f52c392900a2ae508169112e12
zcash_client_sqlite/src/wallet/init/migrations/add_transaction_views.rs	22752	e7497f8b2a80d2ab609c58828acfd3fe99a223a5eaa1a3a82ad559fbde80
zcash_client_sqlite/src/wallet/init/migrations/add_transparent_receiver_address_index.rs	40073	403669054ead8e1900edbf45aa9c19ab2f9de46c7fa20cc2605662b3707a
zcash_client_sqlite/src/wallet/init/migrations/add_transparent_value_index.rs	4433	c24e1a020968b27ce30ab91226ae0ae852b63c59a9588e4c0b50175cb886
zcash_client_sqlite/src/wallet/init/migrations/add_utxo_account.rs	8785	81142244ffd793cc474c6aa4e5fc21897b3659ddcb36c40a27d1c94e505e
zcash_client_sqlite/src/wallet/init/migrations/addresses_table.rs	8586	a2f0a1e56268152437d08cbfa3d9dbf1b9673f48ffb5708484609e18483b
zcash_client_sqlite/src/wallet/init/migrations/ensure_default_transparent_address.rs	1770	53f868550008895da6a567d609be662dddecf418e352dfd560a8ce4cfcd9
zcash_client_sqlite/src/wallet/init/migrations/ensure_orchard_ua_receiver.rs	7050	677cb41c2312f6a684c1b2b2514ce0c7ed273f55af10f566fc5b5a28eb4a
zcash_client_sqlite/src/wallet/init/migrations/ephemeral_addresses.rs	5533	e9a88d7c24b6976f283a60c4cbed50709e86cc68f0624212181a0a2c89a1
zcash_client_sqlite/src/wallet/init/migrations/fix_bad_change_flagging.rs	2537	5fd840389fcfcb88a449471be1a910c552e29eb618a0436927657665d2c3
zcash_client_sqlite/src/wallet/init/migrations/fix_bad_ironwood_change_flagging.rs	22818	4dae69fdfc444c5fe60ecd15f253ccf7d3133fd04c609bf14851bbe7818d
zcash_client_sqlite/src/wallet/init/migrations/fix_broken_commitment_trees.rs	18081	e6f4c290f908a999b604f19e6ee6a144de548db4fafd6b0faa56eabac77e
zcash_client_sqlite/src/wallet/init/migrations/fix_transparent_received_outputs.rs	2995	1cff58dfda8bb0629daf8c0a99a3a75ed75777a716099214a0cd57b6ebec
zcash_client_sqlite/src/wallet/init/migrations/fix_v_transactions_expired_unmined.rs	11430	cea73bf94c60f7b74c7bae313a37b6a80ff698c16bdc0f05b050bcc8c444
zcash_client_sqlite/src/wallet/init/migrations/fix_v_transactions_multi_account_totals.rs	18884	aa4ad6ddcb6679a2a0657db8c65f742459f1554dc2c80bba6475132a70be
zcash_client_sqlite/src/wallet/init/migrations/full_account_ids.rs	29007	605b19ea55f4a4efc7e06ef2822a1744037fd86554a04526b5a21c56e19f
zcash_client_sqlite/src/wallet/init/migrations/initial_setup.rs	4503	add91f083cf39ad1571be95c1753ec0b18adf6e8c6ddfeb8d3fb1d4c73f7
zcash_client_sqlite/src/wallet/init/migrations/ironwood_pool_code_views.rs	14288	041dfed30898dbaf21c37d39e7673840d0389ab09eba1a06ddd3686f0630
zcash_client_sqlite/src/wallet/init/migrations/ironwood_received_notes.rs	4283	94f2b7937b501acc388c826ca5d678920d7fb70137d584bc70ce7104fd35
zcash_client_sqlite/src/wallet/init/migrations/ironwood_shardtree.rs	15048	ff27fa1e55331b5af1955ad5ffd673ebc86c47b9b5c57cb4c55adf1d8543

zcash_client_sqlite/src/wallet/init/migrations/ivk_item_cache.rs	18962	4308a8cc36ddf89eb618053326dae6418a54dc7a4e258261f8e9f596b85c
zcash_client_sqlite/src/wallet/init/migrations/note_locking.rs	3015	8cd352c8fc53308617b4d8675ff0f0ec9acac7e703255e9d2a540d845366
zcash_client_sqlite/src/wallet/init/migrations/nullifier_map.rs	2705	47e97be08acd840447062e20fe9d8e44e71bf3ebcc850e687daccdedc8d3
zcash_client_sqlite/src/wallet/init/migrations/orchard_ironwood_broadcast_binding.rs	10177	a9608a9d73bfe4b89c9e9e6b2b932f381a0151cc2fa2d9173027e0acc926
zcash_client_sqlite/src/wallet/init/migrations/orchard_ironwood_migration_anchor_interval.rs	8593	df6b00200c0adbe1662f831797a3d92e2ab487cb68a702201cc43c85c75a
zcash_client_sqlite/src/wallet/init/migrations/orchard_ironwood_migration_history.rs	8802	fec7e08d071629ab2ac7c989622f8e5049be7d189c2426d054826ab96d97
zcash_client_sqlite/src/wallet/init/migrations/orchard_ironwood_migration_tables.rs	8788	31717b78122ca049ce932f0391d94ff995e6e5851d4fe578ac84b41a80e4
zcash_client_sqlite/src/wallet/init/migrations/orchard_ironwood_migration_txid_blob.rs	10033	44cbd5e0e37643efef1862c03ad49ca06e74d6d4c0b635566b79121830ef
zcash_client_sqlite/src/wallet/init/migrations/orchard_ironwood_migration_unsatisfiability.rs	48868	79372e305f9e85fe0ebbb547eae8a5e071273a0377a55ba68c5d05f8b917
zcash_client_sqlite/src/wallet/init/migrations/orchard_note_version.rs	6786	364a7482f9da35b9b575a8f65079a83ff88a5661e254d78bf210861beaf2
zcash_client_sqlite/src/wallet/init/migrations/orchard_received_notes.rs	15991	e507d78c88f006ab6f93cda9a988cb673603a41e6db3578febc6e72d8d5b
zcash_client_sqlite/src/wallet/init/migrations/orchard_shardtree.rs	10024	6ad70d1ce88834c642c64e9a446c917855e3142d28c405c2579baa0793e9
zcash_client_sqlite/src/wallet/init/migrations/received_notes_nullable_nf.rs	17731	3af819ec77a78081c2b6998c375e8eda9755dfab604a73f34431a6a8b098
zcash_client_sqlite/src/wallet/init/migrations/receiving_key_scopes.rs	35878	da2840dd7c17c3b95a93f46fa9fd8923bcfe683616b5b809dc82d2091661
zcash_client_sqlite/src/wallet/init/migrations/restore_ua_transparent_receivers.rs	19945	acad8bc4a0e1bcb1fd088abade5fa1cebff47219a1d02477bbffe7c3f5fd
zcash_client_sqlite/src/wallet/init/migrations/sapling_memo_consistency.rs	11851	873c8ba60cfccd14f6400fb35844679452a19b4af0218b7baf5b7d7c6780
zcash_client_sqlite/src/wallet/init/migrations/sent_notes_to_internal.rs	2978	3ea5c08d6a54873cd114ffeefee23fadca2866c5d923eb1d222492bf430a
zcash_client_sqlite/src/wallet/init/migrations/shardtree_support.rs	12354	27a78578362e7222817951d292b0fbbbb4072a648716b1b80cc88816f6a0
zcash_client_sqlite/src/wallet/init/migrations/spend_key_available.rs	1545	a94ed1ee3cc1d67812b4290a0e6f88c0309626dc5e825b4039b021be1443
zcash_client_sqlite/src/wallet/init/migrations/standalone_address.rs	12952	9931bdd4d3fe8cbb441e43d41ab4cc732fcc6d3839d42a30f41934f648c3

zcash_client_sqlite/src/wallet/init/migrations/standalone_p2sh.rs	12194	1fedcb2979bde6ebdecdf3c7f78c6043c01c09c37b4ff549ad912d2535be
zcash_client_sqlite/src/wallet/init/migrations/support_legacy_sqlite.rs	3451	5c483f1bb18ddf7dd1c1bb3a0495f27b738d0d96338e2cc31de114dca9f0
zcash_client_sqlite/src/wallet/init/migrations/support_zcashd_wallet_import.rs	4975	5a24aebb41da6c6f285ad2bb641e4e8ddc0dd08fe8703c0160298537383
zcash_client_sqlite/src/wallet/init/migrations/transparent_gap_limit_handling.rs	35549	931eb3f42473d6bc3437f1f03da14e3ccb0a14c28736c8047623c82968f
zcash_client_sqlite/src/wallet/init/migrations/transparent_spend_locator_map.rs	3515	363a1f1b46a5e4cef3163595bde46246e2999cefd338099aa641a422cca
zcash_client_sqlite/src/wallet/init/migrations/tree_retained_checkpoints.rs	2003	2f5136decc9113411340414eb148b1ee0bad42d3063945fc081615d62e57
zcash_client_sqlite/src/wallet/init/migrations/tx_observation_height.rs	4312	71f2e2973f765e100ce1418fb683bc1aadff6da2dcf6479ffea4a57197e8
zcash_client_sqlite/src/wallet/init/migrations/tx_retrieval_queue.rs	15556	bb66209d9c0b85d9b02950fe1af6f1a097fdf081b7e4f4681fec7d5ef4e6
zcash_client_sqlite/src/wallet/init/migrations/tx_retrieval_queue_expiry.rs	2699	aeade75ebff45958f920193199f076f92e34c953a56446f69df832d13dd9
zcash_client_sqlite/src/wallet/init/migrations/tx_status_observation_intent.rs	4837	b7fdd04deff5f19a8c86bd4fe9b9f2ff85b2a9c3cc4e438dcf1a8109692d
zcash_client_sqlite/src/wallet/init/migrations/ufvk_support.rs	13412	378efd2dc5c11da405a0005c3a9aa5ac8cee484d2a21159550e936348ffc
zcash_client_sqlite/src/wallet/init/migrations/utxos_table.rs	1968	640627dfd0d12516cdd6788d5b7c444997bf8f8a1fe5cb4d031ba8777387
zcash_client_sqlite/src/wallet/init/migrations/utxos_to_txos.rs	16710	669feb2065f40eda87a9aa689202de8a039815570d42913168dbc5e2dc41
zcash_client_sqlite/src/wallet/init/migrations/v_address_uses_ironwood.rs	17953	faf213325ae664c1cc93a32557f226c086531eb55e18d5366bc01370306c
zcash_client_sqlite/src/wallet/init/migrations/v_migration_transactions.rs	10722	157015449b027a41435f5718d527f0bee05d31496e629eb35f9218f4feff
zcash_client_sqlite/src/wallet/init/migrations/v_received_output_spends_account.rs	2619	56eba967b26ebc66dcdafb7536ac147a222b904552d2db02970e1860854f
zcash_client_sqlite/src/wallet/init/migrations/v_sapling_shard_unscanned_ranges.rs	4147	033cabe7143f6e0b4de32d4ad137675013d17138ee7feea562b65a82dd2c
zcash_client_sqlite/src/wallet/init/migrations/v_transactions_additional_totals.rs	11460	d08b855526ff81d469ee62750f8853ef918cb1849022ee681cda65e521e4
zcash_client_sqlite/src/wallet/init/migrations/v_transactions_net.rs	26954	77de07654b2f02bd19555191452dc25ab89610223e9f9c8d60ba7ab2c81b
zcash_client_sqlite/src/wallet/init/migrations/v_transactions_note_uniqueness.rs	11721	9b79484fb9d3892e7402b05c8176613f0ecb1f1c4e0663662dac3f9c5e6f

zcash_client_sqlite/src/wallet/init/migrations/v_transactions_pool_crossing.rs	13764	b84fa7018889d024565a0e27f65217e983f87dc1c3d0411bf73eb95ff883
zcash_client_sqlite/src/wallet/init/migrations/v_transactions_shielding_balance.rs	7456	9dc520a6299c90c48b9be0a94cac043f1da7f40b0e56027663024b4d694d
zcash_client_sqlite/src/wallet/init/migrations/v_transactions_transparent_history.rs	9583	1385d9534bee4c63c3e7ba6d8744faf2c73345a5f815e7697d250b9860a5
zcash_client_sqlite/src/wallet/init/migrations/v_transactions_zip318_kind.rs	24512	9f8824f724d8b45a98f4017a52cd63633388b3ee53eefe4d11246134c7cc
zcash_client_sqlite/src/wallet/init/migrations/v_tx_outputs_diversifier_index.rs	7209	96838ec6efd6278b6dd0f06dc8aba8fc823362a305d5245a093af21c866a
zcash_client_sqlite/src/wallet/init/migrations/v_tx_outputs_key_scopes.rs	7438	9de1f632b68aa633ba2bf4ce5dffe4c446210d6ef4e8403581e57b47b0c4
zcash_client_sqlite/src/wallet/init/migrations/v_tx_outputs_return_addrs.rs	4694	c75cffa566544cf90ff8121439845c5615193616c42f9afe730c70c41608
zcash_client_sqlite/src/wallet/init/migrations/v_tx_outputs_transparent_addresses.rs	8108	2f7ccea98408976264155 99bbaf23c9197ea90bf5ba4e348f0543175933a
zcash_client_sqlite/src/wallet/init/migrations/v_tx_outputs_use_legacy_false.rs	4273	5172fcb9f2508f75304c191a4ad4b78e1095ae8648597c5170fae15ed9a2
zcash_client_sqlite/src/wallet/init/migrations/wallet_summaries.rs	3511	de375722dfdad7f620e5de700f1255395fda97cc5ccc6139e45b40cbe7d6
zcash_client_sqlite/src/wallet/init/migrations/witness_stabilized_notes.rs	29059	edd790e468f9fdbfc798d06dd83f5d7a54b196df2fa6c768a139f9800d64
zcash_client_sqlite/src/wallet/init/migrations/zip318_classification.rs	4375	055cf111a46c29af1b26ca308e97e1da3672d7188235cf2991e1a1548442
zcash_client_sqlite/src/wallet/locking.rs	56118	fc5469bec5281ee88a9170bcb7a8c08ba5a7fd491b2aa5f3e07380b9bab3
zcash_client_sqlite/src/wallet/orchard.rs	127938	3c4e95f7dde5d47f588df147e99a6f7e65198a3f3e5cc1a6f832c414e95e
zcash_client_sqlite/src/wallet/sapling.rs	28207	60316cab5d6cd516b120e02f3016dd65a488050010d8db58d35056252816
zcash_client_sqlite/src/wallet/scanning.rs	106495	fecf821f993583cd5dd916291a237e7ba90b19aade4f5ecd6a78bd802db7
zcash_client_sqlite/src/wallet/transparent.rs	214071	84e2f48cd5852a7eab8a7ce8abefda436fe9b4c88357c7ea7b98ec0d557e
zcash_client_sqlite/src/wallet/transparent/ephemeral.rs	8865	1c31f267a81867448f336ff5d2aa95009d133891666a736c62c77c9af303
zcash_client_sqlite/src/zewif.rs	120590	60953a37840964dab7212eff27a39ea44b5d6f0ab5890fa826d88dfc0dfe
zcash_history/Cargo.toml	737	007d33a003baaf37bd7525a23b872279f717f0a63593ff67364b18525933
zcash_history/src/entry.rs	4261	bfc5efe75f619147426eaee8351316702a90ab9c7ab230f17a12e4350f7d
zcash_history/src/lib.rs	2508	6404d6449ef8653ba945532ba3e51985fc80dab3e054598b26dcd09eb547
zcash_history/src/node_data.rs	21997	ef26fa5dec6df0714f93c21c32e4cebe5f7d405c268a578cc3bff394c8c1
zcash_history/src/tree.rs	26295	74630d10192edcaee9beedac18ad92d942b03cd6d7f079938040530b2a03
zcash_history/src/version.rs	6454	08ca0b318412f6f4a5fe776847825e9696e2f28814f4277641770dbd6867

zcash_keys/Cargo.toml	3839	410cc0bcb8164b68f0fb444f66eca1f17bf61f1fc8f04c76d1685624f478
zcash_keys/src/address.rs	50294	52e47576d0a2a94be1b549a55010b4e0ac784cab23d461121340971f394b
zcash_keys/src/constants.rs	5910	3b7682d90f0dfbb8010aba93716e2d7476f7ab00abe7231c3a1575144b19
zcash_keys/src/encoding.rs	25472	2be659e2ef6fc9782ae538c17f52b7ddda7714fe61008d67e7f0255138e3
zcash_keys/src/keys.rs	141099	bd0082831768f382b2fb6d33c41403b7c8730d6e0a2453bf63a7ea741beb
zcash_keys/src/keys/transparent.rs	18488	856dc76b2357dbcf0ab84b3e498152a00571b654bb1f2bd19b9501f8224f
zcash_keys/src/keys/transparent/gap_limits.rs	12352	7a648ae18f271410a5aea6721db339a5a25ab7f537f3b4d9679a6af0e5cc
zcash_keys/src/keys/zcashd.rs	9900	ec4e74fcfee42cc53f9213c854457f9ebf9b5755d8364af5852a0aaf2b1d
zcash_keys/src/lib.rs	752	7dc430b8e92e66884febbff95df313fcf7f2e6cea2f2c2ed0cbeae7ce0d7
zcash_pool_migration/Cargo.toml	4427	84cb4c233c4b280b29b93c2c23d43d05d20ecee6fdd7a3e195301e7f75f0
zcash_pool_migration/src/build.rs	12554	22819e71f25e0bd91ff799c2c8b00c4c917a40bd9603cbf39dfaaabc8e78
zcash_pool_migration/src/denomination.rs	21724	38609669ae371eb784956a7693a2bab0cd86e6326d601fec088707dd9e54
zcash_pool_migration/src/denomination/strategies.rs	47233	a78c67da51660b13097869dd8666491c200384ac76697772cf1cd0f9771f
zcash_pool_migration/src/engine.rs	383516	aee6f68d4539cd01414db1c7427da14c31ac7c63410d233a62a75c39c38b
zcash_pool_migration/src/lib.rs	1710	9b116e1efff4dce35229cad7b271893ee3010bfc1695bc95416c59f9d1ba
zcash_pool_migration/src/pczt_spends.rs	3970	9803b7ee1f23669c57340ec3b2655baff6b78340c4251852e70e2a69f1a6
zcash_pool_migration/src/pczt_txid.rs	3430	10679e58849c4bd6f76b37cb8d2c1e44054f1208d5553d60c8b01cec8725
zcash_pool_migration/src/preparation.rs	44456	41b9d6cce1bae5f5e40d3e20e4bfda2c24b7e0a10a4c60d28f97e79cf869
zcash_pool_migration/src/preparation/first_fit_decreasing.rs	11773	7660be41f417f9f78f852897194fb9ceb934c12cf5893d3e0962178ae8d4
zcash_pool_migration/src/preparation/layered_greedy.rs	53089	b03646d4c86b870fbf0fb3cd9c6a76d1d7485536e5ed449603a06d75611a
zcash_pool_migration/src/satisfiability.rs	146836	575f140a8c640fe9222a4e343c9c28a3bb9300f39432ee7cae3fd547563b
zcash_pool_migration/src/scheduling.rs	113722	8026c5943974240474c06d942bc4c0f9307359945f8039d2b10e274f8654
zcash_pool_migration/src/signing_rounds.rs	42189	3721e27617e658af99ab33823b59f4639a6d8acd3de7eb22f3d23dc3a829
zcash_pool_migration/src/state.rs	218376	33c36ea159ddbce8670ea9729c2e28d466cca2ca5bf27e1aeba18bc96880
zcash_pool_migration/src/wallet.rs	49949	7c8410b69ffcacd4dbb0dd9fcd8855bfa385cf011db35ed5bb3271c030c5
zcash_pool_migration_memory/Cargo.toml	1314	8cbb83b9a2904b50ebaeb70a8507e3b3303c2a9f639e79c62c67fcd1954d
zcash_pool_migration_memory/src/lib.rs	23628	495d726d074fb069a4eb8dc58685612d30b97a707f716216bd9556629279
zcash_primitives/Cargo.toml	4287	0fa139a594a721f7b9e3c74013ffbff8ee76be67347f15f5c99732d63917
zcash_primitives/src/block.rs	23635	10bcceab220ab13b680f2461a27f375b292ad50a76abad9f6b1eeefdbac9
zcash_primitives/src/encoding.rs	2244	ea5912560f2c21a11481392a95a361b2cb0f2160e319c5529034300a6d27
zcash_primitives/src/lib.rs	779	c98e5f8821a7a621efb81686391d09f33a215e7ca5df05972017d2b07611

zcash_primitives/src/merkle_tree.rs	115486	364243f99d68009602c14be8c5543fabb04fa1e49be166e7ce3b8859692b
zcash_primitives/src/transaction/builder.rs	128016	56b71d41eee531b8ddbd13ab856c2d3654b19042051091b364722eb949f6
zcash_primitives/src/transaction/components.rs	228	01ded28e2e67b2a3420dc7b20ad6e785b3e4a77c44e2a976421feb54d9f3
zcash_primitives/src/transaction/components/orchard.rs	24336	bf0f5a8044330d1c8eb55c3fefbe28b66cf3d502222ea1c65a3017158961
zcash_primitives/src/transaction/components/sapling.rs	25048	ec8e4d0d81abadf1d86b01394c00710f27d187c9fd9ef1e794ba8865cf53
zcash_primitives/src/transaction/components/sprout.rs	8128	6b70b0178ea3334a46daf7e6071dfe237b0b91be931be74883a5793010b6
zcash_primitives/src/transaction/fees.rs	1339	e85c578cbc2916338cec03d1a378167d25dfb66003771f1f4b296c215ddf
zcash_primitives/src/transaction/fees/fixed.rs	1231	d95362cfd01c55cce089d826d05f15ea18efd05ae8bf0fd1ab3091485809
zcash_primitives/src/transaction/fees/transparent.rs	3878	531200a6aab1a990c335af28cb880f2751a17132e5d84848440b840cd3a7
zcash_primitives/src/transaction/fees/zip317.rs	8832	45758141236b7745977bd95191dfab619d6588310720652a8af0cbdf0d1d
zcash_primitives/src/transaction/mod.rs	56478	39fcd714c806fcd1b412053604015351817789682 3eca2f94a72a7ec8949
zcash_primitives/src/transaction/sighash.rs	1908	6ee5ca91695de9095fe174265ade901703470a88544fe8825089d1eab585
zcash_primitives/src/transaction/sighash_v4.rs	8975	59239c04ed59d0d5d19c58274524eeb0f588a4f690154d344ae880fa5f72
zcash_primitives/src/transaction/sighash_v5.rs	6221	13155ec1c819691f552d7bc364e3f7bbf24b9d7fa780dd922ddf8f423ab6
zcash_primitives/src/transaction/sighash_v6.rs	1126	76aa108bca4d920e5d889fe22318398b260b15a55ab5979bb71a228de9ba
zcash_primitives/src/transaction/txid.rs	23942	7dd5b8d891270f207d7e6e0aee8681d2d7d38e77b74e9772267ee55c3e1f
zcash_proofs/Cargo.toml	2551	b8a713169619b05777a544e928f72be5caeae02743683b36bd93d8e6c104
zcash_proofs/src/circuit.rs	87	b31c395401ea947e59c9779188267592ff37bf90deed3f92cdf1c3d56c86
zcash_proofs/src/circuit/sprout/commitment.rs	1004	c90b73fdd046ca818bc80b92606e6a5a210e0fa809c09ba09f322655565d
zcash_proofs/src/circuit/sprout/input.rs	5966	f488b72238ef3959b0686b25212ab35d44c050267062416e92f5b41989b4
zcash_proofs/src/circuit/sprout/mod.rs	14049	2ef74af1024fb6cc5584eb781e52016032aa305d09db36950c56ddf39e07
zcash_proofs/src/circuit/sprout/output.rs	1102	6139da53aaf18f813592752f245700acad467354f2482359997fc37f8530
zcash_proofs/src/circuit/sprout/prfs.rs	1939	cb836c15cb412fdc73b77c2a9eb9d91fc6846e20b5438c1545e37ac01b7f
zcash_proofs/src/downloadreader.rs	2893	4ae40df29495f54106dad9bd07d29293d313f0d1cf34964fc6e29dbd675e
zcash_proofs/src/hashreader.rs	1560	ed234ce00866121eba96024416e37341e265727612c46ec111c51e3b7a81
zcash_proofs/src/lib.rs	17442	c2050f2e89d86a3c7a4624a144b88599d2520d68e38e62664c5636ce2a02
zcash_proofs/src/prover.rs	6498	a0242a3ed972b317697275fb508347b681f2c8285f3d8435b36b9b0829ed
zcash_proofs/src/sprout.rs	4687	7b077eeef2f8d854ef8017f66b644ac2272a3031663bd30232b9c6e96d58
zcash_transparent/Cargo.toml	2115	266a1b202974c31e2aa2163d56262f133bbbe04b94d31d0b1f7a289967f2
zcash_transparent/src/address.rs	7530	6a5e646631eaeaa8e301505af6302c5faa36b99aaf5769d3979c8e694728

zcash_transparent/src/builder.rs	43233	21300a7a6424fc10a2c0dda16f6ddac7b295bf72f3e753fc0e48402f8609
zcash_transparent/src/bundle.rs	16351	61a4bd9b83e0d3951d22acc8d4aac8e04c3f27b98fe3e6290eed4794a14d
zcash_transparent/src/coinbase.rs	2327	aa3ba16769a0377c6616e2de244f707a12c3ca5ca20ca71003f902858603
zcash_transparent/src/keys.rs	34597	7f85ab00e7e7ea2e3f53d141d4c6376704c012ccc25a928bb5149ba2745d
zcash_transparent/src/lib.rs	514	6143fcd0e33c0d4978931c1dbe09cfa21d01c33880511a0fd0c31314e833
zcash_transparent/src/pczt.rs	15104	c98366910ea2df8ffbe9356fe62e10d9bdb71a94e535fd98bc93ce376dbd
zcash_transparent/src/pczt/parse.rs	7146	180d4cf4eef89d24e87d30a90fb84f2ca68479eab8c3c0e4152ad8f57f8c
zcash_transparent/src/pczt/signer.rs	22060	7634be395365abd176a608264c15fef69e68584b1c952f6ecca08dacd45c
zcash_transparent/src/pczt/spend_finalizer.rs	17700	2dfb9441411641a48fc898ea0a5548e6a6ae3255e2d195ad9554fd614b29
zcash_transparent/src/pczt/tx_extractor.rs	3441	fbc34ffca7f11fc41da31bf32afc4b87bf26d5eb55d5922961081f5695c0
zcash_transparent/src/pczt/updater.rs	5807	5451e8746f1a5108d7facedd6bb372978c671918c150f9b8ccf9814e5bbc
zcash_transparent/src/pczt/verify.rs	2297	9045b2585a927470ba9502cadb711076af10e2ac1aaa3dedab74a061c427
zcash_transparent/src/sighash.rs	9614	4417b5e6a2cbfb68448fa7fec432b9a30849b78fafdb545ba192a55845a3
zcash_transparent/src/util/hash160.rs	2086	a813a76278afe5125733f3219585270fb88521cfbc061554d8f938fa3e92
zcash_transparent/src/util/mod.rs	94	f59746e68267b6ce74ba40c9971820cfb3384f9884f77a4a25173a122213
zcash_transparent/src/util/sha256d.rs	1609	89f5102feeaddd0b5d4895f563d84997631b36863b2743eb65cb33bce0da
zcash_transparent/src/zip48.rs	35462	cad42458c0611cd3aa21751df669c2b3a89e2c074bbcbc3bba25e65f3c91